Two Hidden Layers Suffice: The Caterpillar Net

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Abstract

Every continuous piecewise linear (CPWL) function is a finite signed sum of maxima of affine functions, making maximum representation a central problem in the study of the expressiveness of ReLU networks. Earlier work disproved the conjectured base-two depth bound, leaving open whether a constant number of hidden layers suffices to represent the maximum. Independently and concurrently, Brunck, Hertrich, Rueß, Stade, and Yehudayoff established that two hidden layers suffice for all CPWL functions. We give the *Caterpillar Net*, a non-recursive explicit construction representing the maximum of N real inputs exactly with two hidden layers of widths $\binom{N}{2} + N + 1$ and $2^{O(N \log N)}$, rational weights, and zero biases. Every expansion of the maximum in its finite vocabulary compiles with the same first hidden layer. Sharing its first hidden layer across a Wang–Sun representation gives two-hidden-layer networks for all CPWL functions. [1]

1 Introduction

In this paper we investigate the representation capabilities of ReLU networks: what functions are representable by such networks of different depths and widths? ReLU networks represent precisely the continuous piecewise-linear (CPWL) functions with finitely many pieces [ABMM18]. A sharp question is therefore: given a CPWL function, what is the smallest depth, or the smallest width at a prescribed depth, needed to represent it exactly [HBD+23, Saf26]? The maximum is a universal building block for such functions: every CPWL function is a finite signed sum of maxima of affine functions [WS05]. Write

$$\max_N(x) := \max_{0 \le i < N} x_i. \tag{1}$$

Theorem (Wang–Sun, [WS05], `Lean_1`)**.** *Let $d \ge 1$ and let $f : \mathbb{R}^d \to \mathbb{R}$ be continuous and affine on each member of a finite polyhedral cover. There are $R \in \mathbb{N}$, affine maps and signs*

$$A_t : \mathbb{R}^d \longrightarrow \mathbb{R}^{d+1}, \qquad \varepsilon_t \in \{-1, 1\} \qquad (1 \le t \le R) \tag{2}$$

such that

$$f(x) = \sum_{t=1}^{R} \varepsilon_t \max_{d+1}(A_t(x)) \qquad \textit{for every } x \in \mathbb{R}^d. \tag{3}$$

[1] Our results were obtained over months of proof search in Lean using Claude Code and Codex; the AI methodology statement describes their roles. The Lean code is available in the accompanying repository https://github.com/JayPiZimmermann/Max_Caterpillar. Links to the Lean formalization appear in theorem titles and equation numbers; equation numbers are clickable.

Thus a two-hidden-layer representation of $\max_{d+1}$ gives one for every CPWL function on $\mathbb{R}^d$: precompose with the affine maps and add the networks in parallel. Section 8 reuses their first hidden layer; the representation theorems and the resulting networks are formalized.

Our contribution. The *Caterpillar Net* gives a nonrecursive two-hidden-layer representation of the maximum with explicit bounds on both hidden-layer widths. The construction uses a finite dictionary of support functions of caterpillar polytopes and their faces. The maximum is the support function of a simplex. We first express this support function as an alternating sum of caterpillar support functions, up to a linear combination of support functions of lower-dimensional polytopes. We then represent the remaining error using faces of the same caterpillar polytopes and elementary ReLU features. Normalizing and merging these descriptions yields a width bound for the complete network, including all correction terms.

Write $\sigma(t) := \max(0, t)$ applied coordinatewise to vectors. A network has an input layer of N coordinates, first and second hidden layers of widths h_1 and h_2, respectively, and a scalar output layer. With real weight matrices W_1, W_2, output vector w_3, and biases b_1, b_2, b_3 of the corresponding dimensions, it evaluates as

$$x \longmapsto w_3^\top \sigma\big(W_2 \sigma(W_1 x + b_1) + b_2\big) + b_3. \tag{4}$$

There are no skip connections; equality is required on all of $\mathbb{R}^N$. For $N = 1, 2, 3$ there are known simple network representations of width $(2, 2)$ and $(4, 2)$ and $(5, 3)$; the last nests $\max(s, t) = t + \sigma(s - t)$. Our Main Theorem presents such constructions for higher N. A network has *rational weights* if every entry of W_1, W_2, and w_3 is rational, and *zero biases* if b_1, b_2, and b_3 vanish.

Theorem 1 (Caterpillar Net, `Lean_2`, Proof)**.** *For every $N \geq 3$, there is a two-hidden-layer ReLU network with rational weights and zero biases computing $\max_N$ with widths*

$$h_1 = \binom{N}{2} + N + 1, \qquad h_2 = 3^{N-2} C_N - N(N-1) + 2, \tag{5}$$

where

$$C_N := \sum_{m=2}^{N-1} \binom{N}{m} \frac{m!}{2} (N - m) m^{N-m-1}. \tag{6}$$

Thus $h_1 = O(N^2)$ and $h_2 = 2^{O(N \log N)}$ as $N \to \infty$.

Further results. The network is compiled from an expansion of $\max_N$ in a finite vocabulary of caterpillar face functions. Theorem 6 separates the network from the coefficients: any such expansion with $|K|$ retained units compiles with the same first hidden layer and $|K| + 2$ second hidden layer neurons. The coefficients are the coordinates of $\max_N$ in one basis of the vocabulary; the same proof shows that they are rational, without any integral (Theorem 11). Section 7 bounds the second width by $q_N + 2$, where q_N is the dimension of the vocabulary modulo its auxiliary functions; this bound proves Theorem 1. For $N = 3, 4$ expansions with one and three units are proved in Lean. Conversely, Section 4 proves that every expansion of $\max_N$ in caterpillar and auxiliary functions has at least $(7/6)^N/(N + 1)$ caterpillar functions, and that $q_N \geq (N - 2)!$. Section 8 applies the construction to sums of maxima.

2 Related work

The conjecture that $\max\{0, x_1, \ldots, x_{2^k}\}$ requires more than k hidden layers [HBD+23] was disproved for $k = 2$ by the maximum of five inputs in [BBH+26]. Exact computer-assisted searches subsequently obtained

two-hidden-layer representations up to $N = 12$ in [RAB+26] and, independently, up to $N = 8$ in [WB26]. The improved general depth bounds remained logarithmic, leaving constant depth open.

Setting all biases to zero does not change a network that computes a positively homogeneous function such as $\max_N$ [HBD+23, Proposition 2.3], [HHL23, Lemma 7], so biases give no additional power for representing the maximum. Depth lower bounds hold for integer weights [HHL23] and rational weights whose denominators are restricted to powers of a fixed integer [AHM25]. Our weights are rational, but their denominators are not restricted in this way, so neither bound applies. Other results concern braid-fan-compatible networks [GHL25] and first hidden layer width bounds [Saf26, Lu26].

Theorem 2 (Brunck, Hertrich, Rueß, Stade, and Yehudayoff, [BHR+26])**.** *Every continuous piecewise affine function $f : \mathbb{R}^d \to \mathbb{R}$ with finitely many affine pieces can be represented exactly by a ReLU network with two hidden layers.*

Their first proof proceeds by induction on dimension, using homological algebra to construct a linear combination of support functions of pyramids over parallelepipeds that agrees with the simplex support function modulo lower-dimensional terms. Our construction restricts these building blocks to an explicitly specified finite family, the caterpillar polytopes; we deliver a geometric proof of the corresponding identity.

Independently and concurrently, they developed a second proof based on combinatorial cancellation, with building blocks closely related to ours. Appendix D identifies them exactly: for nonempty marked sets their pyramids are translated caterpillar polytopes, and the remaining pyramids are rooted stars, which are not caterpillar functions. We do not claim that their coefficients agree with ours. We establish explicit width bounds for the first and second hidden layers; their paper states no widths.

3 The Caterpillar Net

Two identities drive the construction:

$$\max(s, t) = t + \sigma(s - t), \tag{7}$$

$$\sigma(\ell(x)) = \sigma(-\ell(x)) + \ell(x). \tag{8}$$

The first suggests a path with attached leaves. A signed sum of the resulting caterpillar support functions agrees with the maximum modulo lower-dimensional terms; proper faces and linear functions correct the error. The second identity reverses hinges to merge units in Section 6.

Fix $N \geq 3$ and write $x = (x_0, \ldots, x_{N-1}) \in \mathbb{R}^N$.

Definition 3 (Labelled caterpillar, `Lean_3`)**.** A *caterpillar description* is a permutation–set pair

$$\pi \in \mathrm{Sym}\{0, \ldots, N-1\}, \qquad \varnothing \neq S \subseteq \{1, \ldots, r\}, \qquad r := N - 2. \tag{9}$$

Position j carries label $\pi(j+1)$, and $\pi(0)$ is the *apex*. The *spine positions* are $\{0\} \cup S$. The *cut set* S separates positions $j-1$ and j whenever $j \in S$. The last spine position q and the *parent* $p(j)$, the nearest preceding spine position, are

$$q := \max S, \qquad p(j) := \max\big(\{0\} \cup \{i \in S : i < j\}\big) \qquad (1 \leq j \leq r). \tag{10}$$

The parent–child edges form the *labelled caterpillar*: a path through the spine positions with each remaining label attached as a leaf to its parent (Figure 1). The apex is separate from the tree.

Definition 4 (Caterpillar function and face functions, `Lean_4`, `Lean_5`)**.** For labels i, j, for the tree edges, and for the head, put

$$\ell_{ij}(x) := x_i - x_j, \tag{11}$$

$$L_{\pi,S} := \{\ell_{\pi(p(j)+1),\pi(j+1)} : 1 \le j \le r\}, \tag{12}$$

$$h_{\pi,S} := \ell_{\pi(q+1),\pi(0)}. \tag{13}$$

The *edge functions* $L_{\pi,S}$ subtract each child's coordinate from its parent's, one for each of the r tree edges, and the *head function* $h_{\pi,S}$ compares the last spine vertex with the apex. For a linear function h, a finite set L of linear functions, and a *state* $\tau(\ell) \in \{0,\ \ell,\ \sigma \circ \ell\}$ chosen for every $\ell \in L$, define

$$U^{\tau}_{h,L}(x) := \sigma\Big(h(x) + \sum_{\ell \in L} \tau(\ell)(x)\Big). \tag{14}$$

Each edge is omitted, kept linearly, or kept as a hinge. Write $U_{h,L}$ for the all-hinge state, and call (h, L) its *function description*, recording the head and inner hinges. The *caterpillar function* of (π, S) and its *face functions* are

$$U_{\pi,S} := U_{h_{\pi,S},L_{\pi,S}}, \qquad U^{\tau}_{\pi,S} := U^{\tau}_{h_{\pi,S},L_{\pi,S}}. \tag{15}$$

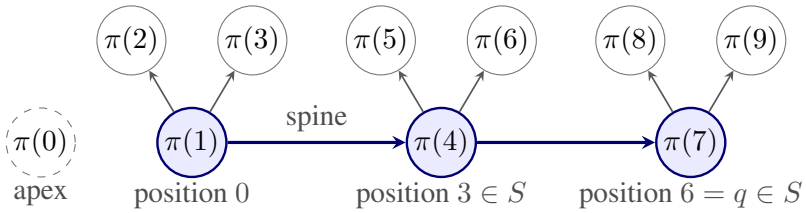


Figure 1: A labelled caterpillar with $N = 10$ and $S = \{3, 6\}$. The spine $\pi(1) \to \pi(4) \to \pi(7)$ has two leaves at each vertex; all arrows point from parent to child. The apex $\pi(0)$ is separate from the tree. The last spine position is $q = 6$, so the head function is $h_{\pi,S}(x) = x_{\pi(7)} - x_{\pi(0)}$.

The term *face function* comes from the following geometry. Identify $a(x) = \langle a, x\rangle$ with its coefficient vector. Then $\ell_{ij} = e_i - e_j$ points from e_j to e_i, and $h = h_{\pi,S} = e_{\pi(q+1)} - e_{\pi(0)}$ points from the apex label to the last spine label. Each $\ell \in L = L_{\pi,S}$ points from child to parent. As free vectors, h places a base corner and the ℓ generate its edges. The polytope apex is 0, distinct from the labelled point $e_{\pi(0)}$. Define the *support function* and the caterpillar polytope by

$$h_P(x) := \max_{p \in P} \langle p, x\rangle, \tag{16}$$

$$Q_{h,L} := \operatorname{conv}\Big(\{0\} \cup \Big(h + \sum_{\ell \in L} [0, \ell]\Big)\Big), \qquad U_{h,L} = h_{Q_{h,L}}. \tag{17}$$

Indeed, the segment $[0, \ell]$ has support function $\sigma \circ \ell$, Minkowski sums add support functions, and adjoining 0 applies the outer ReLU. The base is the translated zonotope $h + \sum_{\ell \in L}[0, \ell]$. For a state τ in Equation (14), set $t_\ell = \mathbb{1}_{\{\tau(\ell) = \ell\}}$, where $\mathbb{1}$ denotes the indicator. It is zero for omitted edges and hinges; the latter contribute segments. The corresponding polytope is $Q^{\tau}_{h,L} := \operatorname{conv}\big(\{0\} \cup (h + \sum_{\ell \in L} t_\ell \ell + \sum_{\tau(\ell) = \sigma \circ \ell}[0, \ell])\big)$, and $U^{\tau}_{h,L} = h_{Q^{\tau}_{h,L}}$. Figure 3 illustrates the translated generators for $N = 4$ on the slice $x_0 = 0$, where a becomes (a_1, a_2, a_3) and e_0 is represented by 0.

The head connects the apex label to the tree, so the head and $N-2$ edge functions are linearly independent. Thus the base is an $(N-2)$-dimensional zonotope with affine hull avoiding 0, and $Q_{h,L}$ is an $(N-1)$-dimensional pyramid. Each nontrivial face containing 0 cones over a base face. Maximizing on each base segment selects 0, ℓ, or $[0, \ell]$, with support functions 0, ℓ, and $\sigma \circ \ell$; the outer ReLU adjoins the apex. These are precisely the three states in Equation (14).

Definition 5 (Dictionary, `Lean_6`)**.** The *dictionary* $\mathcal{D}$ is the finite set of functions $\mathbb{R}^N \to \mathbb{R}$ consisting of every face function $U^\tau_{\pi,S}$, over all caterpillar descriptions (π, S) and all states τ, together with the *auxiliary functions*

$$\begin{aligned} &x \mapsto \sigma(x_i), \quad x \mapsto \sigma(-x_i) \quad (0 \le i < N), \\ &x \mapsto \sigma(x_i - x_j) \qquad\qquad\quad (0 \le i < j < N). \end{aligned} \tag{18}$$

Heads and edge functions are coordinate differences, so face functions and the functions $x \mapsto \sigma(x_i - x_j)$ are translation invariant: with $\mathbf{1} := (1, \dots, 1) \in \mathbb{R}^N$,

$$U^\tau_{\pi,S}(x + t\mathbf{1}) = U^\tau_{\pi,S}(x) \qquad (t \in \mathbb{R}). \tag{19}$$

Only $\sigma(\pm x_i)$ change under this translation. Theorem 10 proves that $\mathcal{D}$ spans the maximum. Section 7 chooses rational coefficients from one basis of its span, without integration.

3.1 The network

The network computes an expansion of $\max_N$ in face functions and auxiliary functions. To compile face functions, first absorb every linear edge of a face state into the head:

$$U^\tau_{h,L} = U_{h',M}, \qquad h' := h + \sum_{\tau(\ell)=\ell} \ell, \qquad M := \{\ell \in L : \tau(\ell) = \sigma \circ \ell\}. \tag{20}$$

By Equation (8), reversing each remaining ℓ_{ij} with $i > j$ gives the same function with the *normalized description*

$$\begin{aligned} M^+ &:= \{\ell_{\min(i,j),\max(i,j)} : \ell_{ij} \in M\}, \\ h^+ &:= h' + \sum_{\ell_{ij} \in M,\ i>j} \ell_{ij}. \end{aligned} \tag{21}$$

Geometrically, $[0, \ell] = \ell + [0, -\ell]$: reversing a generator shifts the head by the old generator. Normalization chooses a consistent orientation and a corresponding base corner; it does not rescale vectors or change the polytope (Figure 3). Write $U_k := U_{h,M}$ for a normalized description $k = (h, M)$, and let $\mathcal{K}_0$ be the set of normalized descriptions of all face functions. Let A_N be the span of the auxiliary functions Equation (18). The network has three parts (Figure 2).

- The *first hidden layer* computes the shared features

$$\Phi(x) := \big(\sigma(x_0),\ \sigma(-x_0),\ (\sigma(x_i - x_0))_{i>0},\ (\sigma(x_i - x_j))_{i<j}\big). \tag{22}$$

- The *second hidden layer* computes the retained units U_k, $k \in K$, for a set $K \subseteq \mathcal{K}_0$, and the two neurons $\sigma(R)$ and $\sigma(-R)$ for an auxiliary remainder $R \in A_N$.
- The *output layer* returns $\sum_{k \in K} d_k U_k + \sigma(R) - \sigma(-R)$.

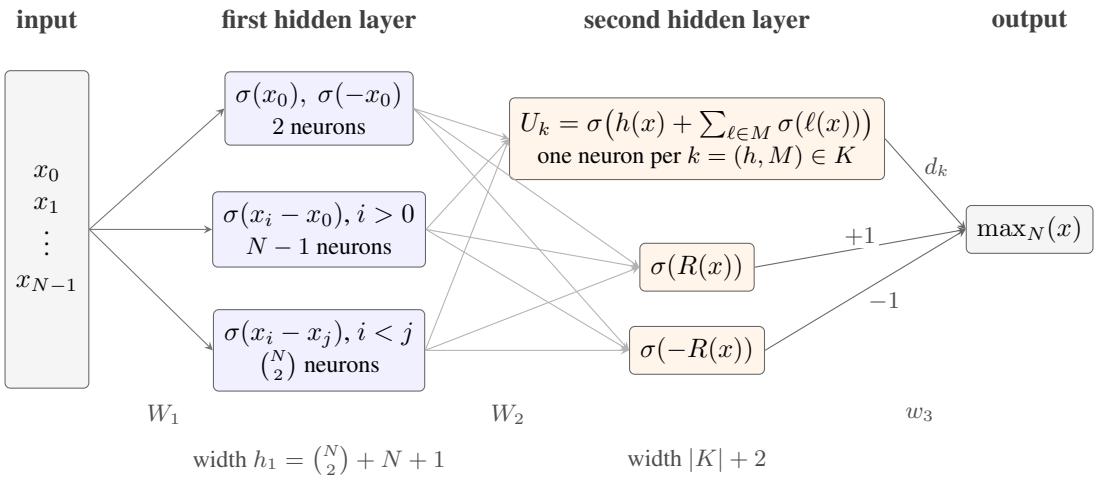


Figure 2: The Caterpillar Net. The first hidden layer is fixed. The second hidden layer has one neuron for every retained normalized description $k = (h, M)$ and two neurons for the auxiliary remainder R. All biases are zero; the matrices are given in Equation (29).

Theorem 6 (Compilation, `Lean_7`, Proof)**.** *Let $K \subseteq \mathcal{K}_0$, $d \in \mathbb{R}^K$, and $R \in A_N$ satisfy*

$$\max_N = \sum_{k \in K} d_k U_k + R. \tag{23}$$

Then the network with first hidden layer Equation (22) *and the matrices in Equation* (29) *computes* $\max_N$*. It has widths h_1 and $|K| + 2$ and zero biases. Its weights are rational whenever the coefficients d_k and the auxiliary coefficients of R are rational.*

Proof of Theorem 6. The first hidden layer outputs Φ from Equation (22). Thus W_1 has rows $e_0^\top$ and $-e_0^\top$, then $e_i^\top - e_0^\top$ for $i > 0$, then $e_i^\top - e_j^\top$ for $i < j$, using a fixed ordering within each block. Its row count is

$$\binom{N}{2} + (N-1) + 2 = \binom{N}{2} + N + 1 = h_1. \tag{24}$$

By Equation (8), these features recover all coordinates,

$$\begin{aligned} x_0 &= \sigma(x_0) - \sigma(-x_0), \\ x_i &= x_0 + \sigma(x_i - x_0) - \sigma(x_0 - x_i) \qquad (i > 0). \end{aligned} \tag{25}$$

For each $k = (h, M) \in K$, substitute Equation (25) into $h(x) + \sum_{\ell \in M} \sigma(\ell(x))$ and collect the coefficients of $\Phi(x)$ into a row vector $w_k^\top$. Every hinge in M already has the orientation $i < j$, so its coefficient is added directly to the corresponding feature.

To specify the remaining row, collect the auxiliary coefficients of R as

$$R(x) = \sum_i p_i \sigma(x_i) + \sum_i n_i \sigma(-x_i) + \sum_{i<j} s_{ij} \sigma(x_i - x_j). \tag{26}$$

Each retained unit is translation invariant by Equation (19), and $\max_N(x + t\mathbf{1}) = \max_N(x) + t$, so Equation (23) gives

$$R(x + t\mathbf{1}) = R(x) + t, \qquad \sum_i p_i = 1. \tag{27}$$

The coefficient sum follows by setting $x = 0$ and $t = 1$. Now choose t large enough that every $x_i + t$ is nonnegative. Substituting the auxiliary expansion into Equation (27) cancels t and gives

$$R(x) = \sum_i p_i x_i + \sum_{i<j} s_{ij}\sigma(x_i - x_j). \tag{28}$$

Substitute Equation (25) into this expression and collect the coefficients of $\Phi(x)$ into w_R. Thus both w_k and w_R are specified by finite sums of the previously defined coefficients. The entries of W_1 and of the rows w_k are integers, because heads and hinges are integer combinations of coordinates. The row w_R is a rational combination of the p_i and s_{ij}, and w_3 consists of the d_k and ± 1.

Order K and set the second hidden layer matrix and output layer vector to

$$W_2 = \begin{pmatrix} (w_k^\top)_{k\in K} \\ w_R^\top \\ -w_R^\top \end{pmatrix}, \qquad w_3 = \begin{pmatrix} (d_k)_{k\in K} \\ 1 \\ -1 \end{pmatrix}, \qquad b_1 = b_2 = b_3 = 0. \tag{29}$$

The second hidden layer outputs $U_k(x)$ and $\sigma(\pm R(x))$. By Equation (8) and Equation (23),

$$w_3^\top \sigma(W_2\Phi(x)) = \sum_{k\in K} d_k U_k(x) + \sigma(R(x)) - \sigma(-R(x)) = \max_N(x). \tag{30}$$

□

Theorem 11 gives such an expansion with rational coefficients; Section 6 counts the normalized descriptions that can occur.

4 Lower bounds for caterpillar expansions

The second width of the network of Theorem 6 is the number of caterpillar functions in an expansion of $\max_N$, plus two. This section shows that this number must grow exponentially, and that the vocabulary itself is large.

Definition 7 (Caterpillar expansion, `Lean_8`)**.** A finite set $K \subseteq \mathcal{K}_0$ is a *caterpillar expansion* of $\max_N$ if there are real numbers d_k, $k \in K$, and a function $R \in A_N$ with $\max_N = \sum_{k\in K} d_k U_k + R$ on $\mathbb{R}^N$. Its *size* is $|K|$.

Theorem 8 (Exponential count, `Lean_9`, `Lean_10`, `Lean_11`, Proof)**.** *Let $N \geq 3$ and let K be a caterpillar expansion of $\max_N$. Then*

$$\binom{N}{m} \leq 6^m\,|K| \qquad (3 \leq m \leq N), \tag{31}$$

$$7^N \leq (N+1)\,6^N\,|K|, \tag{32}$$

$$N(N-1) \leq 6\,|K|. \tag{33}$$

In particular $|K| \geq (7/6)^N/(N+1)$, and every network of the form of Theorem 6 for $\max_N$ has second width at least $(7/6)^N/(N+1) + 2$.

The bound Equation (32) follows from Equation (31) by summing over all m with the binomial theorem; it exceeds the quadratic bound Equation (33) only for N of order 70. The idea of the proof (Appendix C) is local. At a point where $m \geq 3$ coordinates tie at the top and the others lie strictly below in a fixed order, the k-th mixed difference in the tied coordinates annihilates every auxiliary function, which depends on at most two coordinates, and does not annihilate $\max_N$. Hence some U_k with $k \in K$ is nonlinear in all tied coordinates there. The caterpillar structure allows a single U_k to do so only at a fraction $6^m/\binom{N}{m}$ of these points.

Theorem 9 (Vocabulary dimension, `Lean_12`, Proof)**.** *For every $N \geq 3$, the span $V_N := \operatorname{span} \mathcal{D}$ of the dictionary satisfies $q_N := \dim(V_N/A_N) \geq (N-2)!$.*

Thus Theorem 11 cannot give a second width below $(N-2)!+2$; together with Theorem 1, $\log q_N = \Theta(N \log N)$. Neither theorem concerns networks whose second hidden layer uses other functions than caterpillar functions, and neither is a lower bound for general two-hidden-layer networks.

5 Why the dictionary spans the maximum

The maximum is the support function of a simplex. To work in its affine dimension, set

$$\Delta := \operatorname{conv}\{0, e_1, \ldots, e_{N-1}\} \subseteq H_0 := \{x \in \mathbb{R}^N : x_0 = 0\}. \tag{34}$$

For every $x \in \mathbb{R}^N$,

$$\max_N(x) = x_0 + h_\Delta(x - x_0\mathbf{1}). \tag{35}$$

It suffices to prove that h_Δ lies in the span of the restrictions of $\mathcal{D}$ to H_0: the substitution $x \mapsto x - x_0\mathbf{1}$ fixes every face function, sends coordinate features to $\sigma(\pm(x_i - x_0))$, and preserves the dictionary span by Equation (8).

Henceforth all polytopes lie in H_0, and heads and edge functions are identified with their restricted coefficient vectors. Let $\mathcal{L}_k$ be the real span of support functions of polytopes of affine dimension at most k; thus $\mathcal{L}_0$ consists of linear functions. The *exposed face* of P in direction $u \in H_0$ is

$$P^u := \{p \in P : \langle p, u\rangle = h_P(u)\}. \tag{36}$$

The opposite face P^{-u} consists of the minimizers. Define

$$D_u h_P(x) := h_{P^u}(x) - h_{P^{-u}}(x) \qquad (x \in H_0). \tag{37}$$

Here u selects the faces, while x is the support-function input; $P^0 = P$ gives $D_0 = 0$. For a simplex facet normal, D_u subtracts the opposite vertex's linear support function from that of the facet.

For a finite linear combination g of support functions, define its one-sided directional derivative at u by

$$\begin{aligned} \partial_u g(x) &:= \lim_{t \downarrow 0} \frac{g(u+tx) - g(u)}{t}, \\ \partial_u h_P = h_{P^u}, &\qquad D_u g = \partial_u g - \partial_{-u} g. \end{aligned} \tag{38}$$

This makes D_u independent of the support-function decomposition. When $\langle u, \cdot\rangle$ is constant on P, the exposed faces cancel; otherwise both have smaller dimension. Consequently,

$$D_u \mathcal{L}_k \subseteq \mathcal{L}_{k-1} \qquad (k \geq 1). \tag{39}$$

The signed caterpillar sum. We seek a combination that agrees with h_Δ up to a function in $\mathcal{L}_{N-2}$. Adding $x_{\pi(0)}$ makes the caterpillar function the following maximum

$$x_{\pi(0)} + U_{\pi,S}(x) = \max\left\{ x_{\pi(0)},\, x_{\pi(q+1)} + \sum_{\ell \in L_{\pi,S}} \sigma(\ell(x)) \right\}. \tag{40}$$

Averaging over permutations gives the required symmetry

$$F_N(x) := \frac{1}{N!} \sum_{\pi} \sum_{\varnothing \neq S \subseteq \{1,\ldots,r\}} (-1)^{N+|S|} \big(x_{\pi(0)} + U_{\pi,S}(x)\big), \tag{41}$$

The signs cancel the extra exposed-face contributions modulo $\mathcal{L}_{N-3}$, as proved below.

Theorem 10 (Dictionary sufficiency, `Lean_13`, Proof)**.** *For every $N \geq 3$, the dictionary of Definition 5 satisfies*

$$\max_N \in \operatorname{span} \mathcal{D}. \tag{42}$$

Proof of Theorem 10. Define the error by

$$f := F_N|_{H_0} - h_\Delta. \tag{43}$$

Step 10.1 reduces dictionary membership to $f \in \mathcal{L}_{N-2}$; Step 10.2 reduces this to $D_u f \in \mathcal{L}_{N-3}$ for all u. Step 10.3 proves these bounds by cancellation, using induction only when deleting a single exceptional coordinate.

Step 10.1 (Reduce to a lower-dimensional error). Assume temporarily that

$$f = F_N - h_\Delta \in \mathcal{L}_{N-2}. \tag{44}$$

Write $Q_{\pi,S} := Q_{h_{\pi,S},L_{\pi,S}}$. Removing the linear apex terms from Equation (41) gives

$$\frac{1}{N!}\sum_{\pi} \sum_{\varnothing \neq S \subseteq \{1,\dots,N-2\}} (-1)^{N+|S|} h_{Q_{\pi,S}} - h_\Delta \in \mathcal{L}_{N-2}. \tag{45}$$

We correct the error by subtracting derivatives ∂_u until it is linear. Faces of $Q_{\pi,S}$ already have support functions in the dictionary span: apex–base faces give Equation (14), base faces give linear functions plus hinges, and the apex alone gives zero. Coordinate features and Equation (8) represent the linear terms and either hinge orientation. Differentiating h_Δ also produces simplex faces; induction on their vertex count supplies these functions.

Claim 10.1 [Simplex faces, `Lean_14`, `Lean_15`, Proof]. For every nonempty $T \subseteq \{0, e_1, \dots, e_{N-1}\}$, the support function of $\Delta_T := \operatorname{conv} T$ lies in the dictionary span.

Start ($|T| = 1$). The support function of a singleton is linear and lies in the dictionary span.

Step ($|T| \geq 2$). Fix T and assume the claim for all smaller nonempty vertex sets. For any current vertex set A with $|A| \geq 2$, remove $v \in A$ using a direction whose value at v exceeds its common value at the other vertices. Writing $\Delta_A := \operatorname{conv} A$, this gives

$$\begin{aligned} u_v &:= \begin{cases} -\sum_{i=1}^{N-1} e_i & v = 0, \\ v & v \neq 0, \end{cases} \\ (\Delta_A)^{u_v} &= \{v\}, \qquad (\Delta_A)^{-u_v} = \Delta_{A\setminus\{v\}}, \\ D_{u_v} h_{\Delta_A}(x) &= \langle v, x\rangle - h_{\Delta_{A\setminus\{v\}}}(x). \end{aligned} \tag{46}$$

Order the vertices outside T as $v_1, \dots, v_{N-|T|}$ and set

$$\varepsilon_T := D_{u_{v_{N-|T|}}} \cdots D_{u_{v_1}} \left(\frac{1}{N!}\sum_{\pi} \sum_{\varnothing \neq S \subseteq \{1,\dots,N-2\}} (-1)^{N+|S|} h_{Q_{\pi,S}} - h_\Delta \right). \tag{47}$$

Each deletion changes the sign of the remaining simplex term, and subsequent D_u annihilate the linear terms. Hence

$$D_{u_{v_{N-|T|}}} \cdots D_{u_{v_1}} h_\Delta = (-1)^{N-|T|} h_{\Delta_T} + \ell_T, \tag{48}$$

where ℓ_T is linear. Thus the coefficient of h_{Δ_T} in ε_T is $(-1)^{N-|T|+1}$; all other terms are linear or caterpillar-face support functions and lie in $\operatorname{span} \mathcal{D}$. By Equation (39), $\varepsilon_T \in \mathcal{L}_{|T|-2}$. For the full vertex set the empty composition is the identity.

Lower the residual dimension recursively by

$$\begin{aligned} \varepsilon_T^{(0)}(x) &:= \varepsilon_T(x), \\ \varepsilon_T^{(j+1)}(x) &:= \varepsilon_T^{(j)}(x) - \sum_{u \in E_{T,j}} \big(\partial_u \varepsilon_T^{(j)}\big)(x). \end{aligned} \tag{49}$$

Here $0 \le j < |T| - 2$; the next claim chooses each finite $E_{T,j} \subseteq H_0$.

Claim 10.2 [Residual reduction, `Lean_16`, Proof]. The sets $E_{T,j}$ can be chosen successively so that, for $0 \le j \le |T| - 2$,

$$\begin{aligned} \varepsilon_T^{(j)} &\in \mathcal{L}_{|T|-2-j}, \\ \varepsilon_T - \varepsilon_T^{(j)} &\in \operatorname{span} \mathcal{D}. \end{aligned} \tag{50}$$

Each residual remains a linear combination of h_{Δ_T}, support functions of its proper faces and of caterpillar faces, and linear functions.

The proof in Appendix A.1 chooses a direction orthogonal to each maximal-dimensional direction space, avoiding orthogonality to the simplex and other spaces. The derivative sum preserves each maximal term exactly once; subtracting it lowers the dimension.

At stage $j = |T| - 2$, the error is linear, so Equation (50) gives $\varepsilon_T \in \operatorname{span} \mathcal{D}$. Its simplex coefficient is ± 1, and its other terms already lie in the span; hence $h_{\Delta_T} \in \operatorname{span} \mathcal{D}$. Taking the full vertex set proves $h_\Delta \in \operatorname{span} \mathcal{D}$ under assumption Equation (44).

Step 10.2 (Why the facet bounds imply the class identity). For $N \ge 4$ and $f = F_N - h_\Delta$, we show that $f \in \mathcal{L}_{N-2}$ follows from

$$D_u f \in \mathcal{L}_{N-3} \qquad \text{for every } u \in H_0. \tag{51}$$

The base zonotope admits the Freudenthal–Kuhn triangulation [Fre42, Kuh60]; coning it to the apex expresses each caterpillar support function as a sum of simplex support functions modulo $\mathcal{L}_{N-2}$. For a simplex, D_u isolates a facet up to the linear support function of its opposite vertex. Thus Equation (51) controls the weighted sums of parallel facets. A common linear reconstruction operator turns these facet cancellations back into a cancellation of the original simplex sum. Define

$$(\mathcal{T} g)(x) := \begin{cases} g(x) & x_1 \ge 0, \\ g(x - x_1 e_1) & x_1 < 0. \end{cases} \tag{52}$$

Claim 10.3 [Simplex reconstruction, `Lean_17`, Proof]. Every $(N-1)$-dimensional simplex $P \subseteq H_0$ satisfies

$$h_P - \mathcal{T}\left(\sum_{F \text{ facet of } P} \operatorname{sgn}\langle \nu_F, e_1 \rangle \, h_F \right) \in \mathcal{L}_{N-2}. \tag{53}$$

Here ν_F is an outward normal of F. Its sign against e_1 records whether the facet bounds lines parallel to e_1 from above or below; parallel facets contribute zero. The operator $\mathcal{T}$ fills the segments between a polytope and its projection onto $p_1 = 0$, up to a segment correction for translation; hence $\mathcal{T}\mathcal{L}_k \subseteq \mathcal{L}_{k+1}$. Applying the claim with the same operator to every simplex preserves the cancellations between parallel facets. Their sums lie in $\mathcal{L}_{N-3}$, so their reconstructions lie in $\mathcal{L}_{N-2}$, as required. Appendix A.2 proves the claim and the subdivision and grouping steps.

Claim 10.4 [Class identity, `Lean_18`, Proof]. The class identity Equation (44) holds for every $N \geq 3$.

Step 10.3 (Prove the class identity by induction).

Start ($N = 3$). Here $S = \{1\}$, $q = 1$, and $p(1) = 0$, so Equation (40) gives

$$\begin{aligned} x_{\pi(0)} + U_{\pi,\{1\}}(x) &= \max\bigl\{x_{\pi(0)}, x_{\pi(2)} + \sigma(x_{\pi(1)} - x_{\pi(2)})\bigr\} \\ &= \max\{x_{\pi(0)}, x_{\pi(1)}, x_{\pi(2)}\} = \max_3(x). \end{aligned} \tag{54}$$

All six terms have coefficient $1/6$, so

$$F_3 = \max_3 . \tag{55}$$

On H_0, this gives $f = 0 \in \mathcal{L}_1$.

Step ($N \geq 4$). Assume the class identity for $N - 1$. By Step 10.2, it suffices to prove $D_u f \in \mathcal{L}_{N-3}$ for every $u \in H_0$. The linear apex terms disappear. The exposed simplex faces have the largest and smallest coordinate values u_i, including $u_0 = 0$. In a caterpillar base, $[0, \ell_{ij}]$ remains free when $u_i = u_j$; its maximizing endpoint is ℓ_{ij} when $u_i > u_j$ and 0 when $u_i < u_j$. Minimization reverses these choices.

Discarding contributions in $\mathcal{L}_{N-3}$. Fix $u \in H_0 \setminus \{0\}$, since $D_0 f = 0$, and a summand $Q_{\pi,S}$. A face lying entirely in its base $B_{\pi,S} := h_{\pi,S} + \sum_{\ell \in L_{\pi,S}} [0, \ell]$, without the apex 0, is a translated sum of segments; its support function lies in $\mathcal{L}_1 \subseteq \mathcal{L}_{N-3}$. The singleton apex has zero support function. If at least two tree edges have endpoint labels i, j with $u_i \neq u_j$, at most $N - 4$ segments remain free; adjoining the apex gives dimension at most $N - 3$. Only apex–base faces with at most one tree edge whose endpoint coordinate values differ remain.

For a face exposed in direction $-u$, replace u by $-u$; we can then always analyze maximizers of $\langle u, \cdot \rangle$. Put $a := u_{\pi(0)}$. The face contains both the apex and base points exactly when the maximum over the base is zero. There are three cases.

Case 1: one spine edge with unequal endpoint values. Removing this edge splits the tree into two components with constant coordinate values $c \neq d$. The maximum over the base is $\max\{c, d\} - a$, so the required condition is

$$a = \max\{c, d\}. \tag{56}$$

The apex label therefore adds no new coordinate value, and u has exactly two distinct coordinate values.

Case 2: one leaf edge with unequal endpoint values. Let d be the leaf's coordinate value and $c \neq d$ the common value at all other tree vertices. The maximum over the base is $c + \sigma(c - d) - a$, so the required condition is

$$a = c + \sigma(c - d) = \begin{cases} c, & d > c, \\ 2c - d, & d < c. \end{cases} \tag{57}$$

If $d > c$, the apex label has value c, giving exactly two distinct coordinate values. If $d < c$, put $t := c - d > 0$. The leaf has value $c - t$, the apex label has value $c + t$, and all other $N - 2$ labels have value c. Up to a permutation of labels, this gives the three-value pattern $(c + t, c - t, c, \dots, c)$.

Case 3: equal values at every tree-edge endpoint pair. If $u_i = u_j$ for the endpoint labels of every tree edge, all tree values equal some c. The maximum over the base is $c - a$, so we would also need $a = c$, making every coordinate of u equal. Since $u \in H_0$, this would force $u = 0$, which we excluded.

Negating u preserves these patterns. Since $Q_{\pi,S}$ was arbitrary, only directions with two distinct coordinate values or the pattern $(c + t, c - t, c, \dots, c)$ can contribute to the full signed sum.

Cancellation for three distinct values. Let α, β be the exceptional labels, with $u_\alpha = c + t$, $u_\beta = c - t$, and $u_i = c$ for every other label, where $t \neq 0$. Let τ exchange α and β and fix all other labels. Define the linear map $R : H_0 \to H_0$ explicitly by

$$(Rx)_i := x_{\tau(i)} - x_{\tau(0)} \qquad (0 \le i < N,\ x \in H_0). \tag{58}$$

Thus R first swaps the two exceptional coordinates and then subtracts the new zeroth coordinate from every coordinate; in particular, $(Rx)_0 = 0$. Since $u_0 = 0$, this formula also gives $Rx = x - \frac{x_\alpha - x_\beta}{t} u$. With the explicit linear functional $\langle b, x\rangle := (x_\alpha - x_\beta)/(2t)$, we therefore have $Rx = x - 2\langle b, x\rangle u$, $\langle b, u\rangle = 1$, and $Ru = -u$. Hence R is a rank-one reflection. Permutation symmetry and the common coordinate shift $-x_{\tau(0)}$ give $F_N(Rx) = F_N(x) + \ell(x)$ for a linear function ℓ. Write $G := F_N|_{H_0}$. Applying the reflection identity to $u + \varepsilon x$ and to u, subtracting, dividing by $\varepsilon > 0$, and taking $\varepsilon \downarrow 0$ gives $\partial_{-u} G(Rx) = \partial_u G(x) + \ell(x)$, because $R(u + \varepsilon x) = -u + \varepsilon Rx$. To remove R from the derivative argument, positive homogeneity gives the identity $\partial_v G(x + \lambda v) = \partial_v G(x) + \lambda G(v)$ for every real λ. Indeed, for sufficiently small $\varepsilon > 0$, write $v + \varepsilon(x + \lambda v) = (1 + \varepsilon\lambda)\big(v + \frac{\varepsilon}{1+\varepsilon\lambda} x\big)$ and apply the definition Equation (38). Using this identity with $v = -u$ and $\lambda = 2\langle b, x\rangle$, we obtain

$$\begin{aligned} \partial_u G(x) + \ell(x) &= \partial_{-u} G(Rx) \\ &= \partial_{-u} G\big(x + 2\langle b, x\rangle(-u)\big) \\ &= \partial_{-u} G(x) + 2\langle b, x\rangle G(-u), \\ D_u G(x) &= \partial_u G(x) - \partial_{-u} G(x) = 2F_N(-u)\langle b, x\rangle - \ell(x). \end{aligned} \tag{59}$$

The right-hand side is linear in x, so

$$D_u(F_N|_{H_0}) \in \mathcal{L}_0. \tag{60}$$

The two exposed simplex faces are vertices, so their support functions are linear as well.

Handling two distinct values. First suppose that each value occurs at least twice. After the preceding dimension bound, every term still requiring cancellation has a unique split $u_{\pi(1)} = \cdots = u_{\pi(k)} \neq u_{\pi(k+1)} = \cdots = u_{\pi(r+1)}$ with $k \in S$. If the apex label has the latter value, toggle membership of $k - 1$ in S. Otherwise, exchange the equal-valued labels $\pi(k+1), \pi(k+2)$ and toggle membership of $k+1$. Repetition of both values gives $k \ge 2$ in the first case and $k + 2 \le r + 1$ in the second, so the operations are defined; the unchanged cut at k keeps the cut set nonempty. The transformed term remains in the same case, and repeating the operation returns the original term. If Φ denotes this involution on the active terms and $w_{\pi,S} := (-1)^{N+|S|}/N!$, then

$$w_{\Phi(\pi,S)} = -w_{\pi,S}, \qquad D_u h_{Q_{\pi,S}} - D_u h_{Q_{\Phi(\pi,S)}} \in \mathcal{L}_1. \tag{61}$$

Inactive fixed points of the global transformation are already in $\mathcal{L}_{N-3}$ by the dimension bound. Summing the active pairs therefore puts $D_u F_N$ in $\mathcal{L}_{N-3}$. Each exposed simplex face has at most $N - 2$ vertices and hence dimension at most $N - 3$, so $D_u f \in \mathcal{L}_{N-3}$ without using induction.

One exceptional coordinate: the induction hypothesis. Suppose one of the two values occurs only once. Choose τ with $\tau(0)$ the exceptional label. To stay in H_0, relabel and subtract the new zeroth coordinate:

$$B_\tau : H_0 \longrightarrow H_0, \qquad (B_\tau x)_i := x_{\tau(i)} - x_{\tau(0)} \qquad (0 \le i < N). \tag{62}$$

Its inverse is $B_{\tau^{-1}}$, and it sends u to $(0, t, \dots, t)$, $t \neq 0$. Transforming both input and direction preserves every $\mathcal{L}_k$ and changes f only by a linear term, annihilated by D. Rescaling, and reversing the sign of D_u if needed, reduces to $u = (0, 1, \dots, 1)$.

This direction exposes the $(N-2)$-dimensional facet $\Delta' := \mathrm{conv}\{e_1, \dots, e_{N-1}\}$ and the opposite vertex 0. Hence

$$D_u h_\Delta(x) = h_{\Delta'}(x) = \max\{x_1, \dots, x_{N-1}\} \qquad (x \in H_0). \tag{63}$$

Here is the exact use of the induction hypothesis. Set $z := (x_1, \dots, x_{N-1})$ and $y := z - x_1\mathbf{1} = (0, x_2 - x_1, \dots, x_{N-1} - x_1)$, where $\mathbf{1} \in \mathbb{R}^{N-1}$ has every coordinate equal to 1. Every caterpillar maximum increases by a common added constant. Averaging over the $(N-1)!$ lower label orders cancels the factorial in the total signed coefficient, giving

$$F_{N-1}(z) - \max_{N-1}(z) = F_{N-1}(y) - \max_{N-1}(y) + (W_{N-1} - 1)x_1 \tag{64}$$

for

$$W_{N-1} := \sum_{\varnothing \neq T \subseteq \{1,\dots,N-3\}} (-1)^{N-1+|T|}. \tag{65}$$

Unlike z, the vector y has zeroth coordinate zero, so the $(N-1)$-input induction hypothesis applies to it. Linear precomposition preserves $\mathcal{L}_{N-3}$, and the last term is linear. Thus

$$\big(x \mapsto F_{N-1}(x_1, \dots, x_{N-1}) - \max\{x_1, \dots, x_{N-1}\}\big) \in \mathcal{L}_{N-3}. \tag{66}$$

Claim 10.5 [Coordinate deletion, `Lean_19`, Proof]. For $N \ge 4$,

$$D_{(0,1,\dots,1)}(F_N|_{H_0}) - \big(x \mapsto F_{N-1}(x_1, \dots, x_{N-1})\big) \in \mathcal{L}_{N-3}. \tag{67}$$

The proof is in Appendix A.3. Its mechanism is that only degree-one tree positions of label 0 survive. Each nonempty lower cut set has N insertion preimages, each contributing $1/N$ of its lower coefficient. The extra empty-cut star sum lies in $\mathcal{L}_{N-3}$ by Boolean cancellation.

Completion of the induction. Combining Equation (63), Equation (66), and Claim 10.5 gives

$$\begin{aligned} D_u f(x) = {} & \big[D_u(F_N|_{H_0})(x) - F_{N-1}(x_1, \dots, x_{N-1})\big] \\ & + \big[F_{N-1}(x_1, \dots, x_{N-1}) - \max\{x_1, \dots, x_{N-1}\}\big]. \end{aligned} \tag{68}$$

Both brackets lie in $\mathcal{L}_{N-3}$, the first by Claim 10.5 and the second by induction. Step 10.2 now gives $f \in \mathcal{L}_{N-2}$, and Step 10.1 gives the dictionary representation. □

6 Network size

The widths in Theorem 1 are worst-case upper bounds: we count all possible descriptions after collecting linear terms into the head and orienting the remaining hinges consistently. Discarding zero aggregate coefficients can give a smaller network; zero padding realizes the stated widths.

The normalized description Equation (21) of a caterpillar function depends only on its undirected spine, apex, and leaf attachments. For m spine vertices these choices give the summand of C_N in Equation (6). Each of the $N-2$ generators has three face states, giving at most $3^{N-2}C_N$ descriptions. The $N(N-1)$ descriptions with no retained hinge and head $x_i - x_j$, $i \neq j$, can be combined with the auxiliary functions. By Theorem 6, the second width is the number of retained descriptions plus two; Appendix B gives the complete count.

Formalization (`Lean_20`). The theorem and equation links point to the Lean development at a fixed repository revision. Appendix E prints the network definitions and the main theorem, and describes the verification certificate.

7 Compression

Theorem 6 reduces the second width to the number of retained units in an expansion. This section bounds that number through the dimension of the vocabulary. Let $V_N := \operatorname{span} \mathcal{D}$, and let $A_N \subseteq V_N$ be the span of the auxiliary functions Equation (18). Put

$$q_N := \dim\big(V_N/A_N\big) = \dim V_N - \dim A_N. \tag{69}$$

Theorem 11 (Quotient compression, `Lean_21`, Proof)**.** *For every $N \geq 3$ there are a set $K \subseteq \mathcal{K}_0$ with $|K| \leq q_N$, rational coefficients d_k, and $R \in A_N$ with rational auxiliary coefficients such that $\max_N = \sum_{k\in K} d_k U_k + R$. Hence $\max_N$ is computed by a network with rational weights, zero biases, and widths*

$$h_1 = \binom{N}{2} + N + 1, \qquad h_2 = q_N + 2. \tag{70}$$

Proof of Theorem 11. Choose a basis of A_N among the auxiliary functions and extend it by face functions to a basis of V_N. The extension adds exactly q_N face functions. For each of them select one normalized description, rather than splitting its coefficient across all descriptions of the same function; this gives K with $|K| \leq q_N$. Let $S = \{g_1, \dots, g_m\}$ be the resulting basis of V_N. By Theorem 10, $\max_N \in V_N$, so $\max_N = \sum_i c_i g_i$ with unique real coordinates c_i.

The coordinates are rational. Every g_i and $\max_N$ is continuous and takes rational values on $\mathbb{Q}^N$, being built from σ and integer combinations of coordinates. A combination of the g_i that vanishes on the dense set $\mathbb{Q}^N$ vanishes everywhere, so the rational rows $(g_i(y))_i$, $y \in \mathbb{Q}^N$, span $\mathbb{Q}^m$; otherwise a nonzero rational vector would be orthogonal to all of them. Hence there are points $y_1, \dots, y_r \in \mathbb{Q}^N$ and rationals a_{jk} with $\sum_k a_{jk} g_i(y_k) = \delta_{ij}$, and

$$c_j = \sum_i c_i \sum_k a_{jk} g_i(y_k) = \sum_k a_{jk} \max_N(y_k) \in \mathbb{Q}. \tag{71}$$

Theorem 6 compiles the expansion; its weights are rational because the c_j are. □

Every face function is the unit of a normalized description outside $\mathcal{P}$ or an auxiliary combination (Appendix B), so $q_N + 2$ never exceeds the second width of Theorem 1. The quotient dimension controls only this compression. A network needs only the units of one expansion of $\max_N$; Theorem 12 gives expansions with one and three units for $N = 3, 4$.

Theorem 12 (Small arities, `Lean_22`, `Lean_23`)**.** *For all real inputs,*

$$\max_3(x) = x_1 + \sigma\big(x_2 - x_1 + \sigma(x_0 - x_2)\big), \tag{72}$$

$$\begin{aligned}\max_4(x) = x_1 &- \sigma(x_0 - x_2) - \sigma\big(x_1 + x_2 - x_0 - x_3 + \sigma(x_0 - x_1)\big) \\ &+ \sigma\big(x_1 + x_2 - x_0 - x_3 + \sigma(x_0 - x_1) + \sigma(x_0 - x_2)\big) \\ &+ \sigma\big(x_3 - x_1 + \sigma(x_0 - x_2) + \sigma(x_2 - x_3)\big).\end{aligned} \tag{73}$$

Hence the network of Theorem 6 computes $\max_3$ *with widths* $(7, 3)$ *and* $\max_4$ *with widths* $(11, 5)$*, with rational weights and zero biases.*

Each outer unit in Equations (72) and (73) is a normalized face function. Lean proves both identities by exhaustive case analysis of the activation pattern on all of $\mathbb{R}^N$. For these arities, smaller networks are known: nesting $\max(s, t) = t + \sigma(s - t)$ gives widths $(5, 3)$ for $\max_3$ and $(6, 3)$ for $\max_4$. The identities illustrate how sparse the expansion in the vocabulary can be.

8 CPWL functions

By the Wang–Sun theorem, every CPWL function on $\mathbb{R}^d$ is a signed sum of maxima of $d + 1$ affine functions. These affine functions can be taken from any finite family among which the function selects pointwise, without rescaling them.

Theorem (Koutschan–Moser–Ponomarchuk–Schicho, [KMPS23, Theorem 2.4], `Lean_24`, `Lean_25`)**.** *Let* $f : \mathbb{R}^d \to \mathbb{R}$ *be continuous, and let* $g_1, \dots, g_n : \mathbb{R}^d \to \mathbb{R}$ *be affine with* $f(x) \in \{g_1(x), \dots, g_n(x)\}$ *for every* x*. Then there are* $T \in \mathbb{N}$*, signs* $\varepsilon_t \in \{-1, 1\}$*, and maps* $\alpha_t : \{0, \dots, d\} \to \{1, \dots, n\}$ *with*

$$f(x) = \sum_{t=1}^{T} \varepsilon_t \max_{d+1}\big(g_{\alpha_t(0)}(x), \dots, g_{\alpha_t(d)}(x)\big) \qquad \text{for every } x \in \mathbb{R}^d. \tag{74}$$

The affine functions of every polyhedral description of a CPWL function satisfy the hypothesis. The Lean proof follows Ovchinnikov's max–min form $f = \max_S \min_{j \in S} g_j$, expands it by inclusion–exclusion into an integer combination of maxima over subsets of $\{g_1, \dots, g_n\}$, and reduces each maximum to subsets of at most $d + 1$ functions by an affine dependence among $d + 2$ gradients.

Reusing the first hidden layer (`Lean_26`, `Lean_27`). Let $d \geq 2$, and let (h_1, h_2) be the widths of Theorem 1 for $N = d + 1$. Substituting the arguments of each term of Equation (74) into the Caterpillar Net computes f. All terms use only the features $\sigma(g_a - g_b)$ for $a < b$, $\sigma(\pm g_1)$, and $\sigma(g_a - g_1)$, so one first hidden layer of width $\binom{n}{2} + n + 1$ serves every term. Terms with the same argument set $\alpha_t(\{0, \dots, d\})$ compute the same maximum and are merged, and the auxiliary remainders of all terms form one pair $\sigma(\pm R)$. The second hidden layer has width $s(h_2 - 2) + 2$, where s is the number of distinct argument sets and $s \leq \min\big(T, \sum_{m \leq d+1} \binom{n}{m}\big)$. Thus the second hidden layer holds one copy of the Caterpillar Net units for every distinct maximum; only the first hidden layer and the remainder pair are genuinely shared. The second hidden layer and output layer have rational weights and zero biases, and every first hidden layer neuron applies σ to $\pm(g_a - g_b)$ or $\pm g_a$. Hence rational g_i give rational weights and biases, and linear g_i give zero biases.

9 Conclusion

The second width of the Caterpillar Net is the number of caterpillar functions in an expansion of $\max_N$ plus two. For $N = 3, 4$, one and three functions suffice (Theorem 12); for general N, Theorem 11 retains at most q_N of them. Conversely, every such expansion needs at least $(7/6)^N/(N+1)$ caterpillar functions (Theorem 8), so caterpillar building blocks cannot give subexponential second widths. Whether the maximum has two-hidden-layer representations of subexponential width with other second-layer functions remains open.

A Supporting proofs for the dictionary theorem

A.1 Residual reduction

We prove the inner induction used in the simplex-face argument of Section 5. Fix a nonempty vertex set T with $|T| \geq 2$, and assume that every proper simplex face on a subset of T is already in the dictionary span, as in the outer induction there.

Proof of Claim 10.2. We argue by induction on j.

Start ($j = 0$). By definition, $\varepsilon_T^{(0)} = \varepsilon_T$. Its dimension bound and support-function expression were established above, and the removed part is zero.

Step ($j < |T| - 2$). Assume the directions for the first j stages have been chosen and the claim holds at stage j. Put $k := |T| - 2 - j \geq 1$. We require each new direction u to be nonorthogonal to the simplex direction space $\operatorname{dir} \Delta_T := \operatorname{span}\{v - w : v, w \in T\}$. Its values on T are then not all equal, so the maximizing vertices form a nonempty proper subset:

$$\begin{aligned} A_u &:= \{v \in T : \langle u, v\rangle = h_{\Delta_T}(u)\}, \qquad \varnothing \neq A_u \subsetneq T, \\ \partial_u h_{\Delta_T} &= h_{\Delta_{A_u}} \in \operatorname{span} \mathcal{D}. \end{aligned} \tag{75}$$

The last membership uses the outer induction hypothesis, since $|A_u| < |T|$. An orthogonal direction would leave h_{Δ_T} unchanged and would not allow this use of the hypothesis.

Choose an auxiliary expression for $\varepsilon_T^{(j)}$ by support functions of polytopes of dimension at most k, and collect their distinct k-dimensional direction spaces. For each such space V, choose $u_V \in V^\perp$ not orthogonal to $\operatorname{dir} \Delta_T$ or to any other collected space V'. This is possible because $\dim \Delta_T = |T| - 1 > k$: each of $\operatorname{dir} \Delta_T$ and V' contains a vector outside V, whose nonzero projection onto $V^\perp$ is not orthogonal to that vector. Adding sufficiently small multiples of these projections avoids each orthogonality condition in turn while preserving those already avoided. Choose $E_{T,j} := \{u_V : V \text{ is a collected direction space}\}$, with $E_{T,j} = \varnothing$ if there are none. In the sum of derivatives over $E_{T,j}$, each term coming from a k-dimensional polytope is preserved exactly once. Every other derivative of that term is the support function of a proper face, while terms of lower dimension remain in $\mathcal{L}_{k-1}$. Subtracting this derivative sum therefore cancels all k-dimensional terms. By the recursive definition Equation (49), the resulting error satisfies $\varepsilon_T^{(j+1)} \in \mathcal{L}_{k-1} = \mathcal{L}_{|T|-3-j}$.

Now compute these same derivatives using the support-function expression in the claim. The derivative of h_{Δ_T} lies in the dictionary span by Equation (75). Differentiating the support function of an already proper simplex face still gives a support function of a proper face of Δ_T, so the same induction hypothesis applies. Differentiating the support function of a face of $Q_{\pi,S}$ gives a support function of a face of the same polytope, whose support functions lie in the dictionary span, and linear functions remain linear. Thus the derivative sum lies in the dictionary span. By Equation (49), $\varepsilon_T - \varepsilon_T^{(j+1)}$ is the sum of the previously removed part and this derivative sum, so it also lies in the dictionary span. Subtracting the derivative sum preserves the stated types of support-function terms. This completes the proof step.

□

A.2 Reconstruction from opposite exposed faces

We justify Step 10.2: for $N \geq 4$, the bounds Equation (51) imply $f \in \mathcal{L}_{N-2}$. We first subdivide the caterpillar polytopes, then prove Claim 10.3, and finally apply it to the resulting weighted simplex sum.

The base of each $Q_{\pi,S}$ is an affine cube. Order its edge vectors by a bijection $\omega : \{1, \dots, N-2\} \to L_{\pi,S}$, add them one at a time, and adjoin the apex:

$$K_{\pi,S,\omega} := \operatorname{conv}\left(\{0\} \cup \left\{h_{\pi,S} + \sum_{a=1}^{k} \omega(a) : 0 \le k \le N-2\right\}\right). \tag{76}$$

These $(N-2)!$ simplices are obtained by coning the *Freudenthal–Kuhn triangulation* of the base to the apex [Fre42, Kuh60]: in cube coordinates, each base simplex occupies one ordering region $1 \ge t_{\omega(1)} \ge \cdots \ge t_{\omega(N-2)} \ge 0$. For a hyperplane cut $P = P_+ \cup P_-$, the support-function identity

$$h_P + h_{P_+ \cap P_-} = h_{P_+} + h_{P_-} \tag{77}$$

shows that subdivision changes the sum of support functions only by support functions of the cutting sections. Hence

$$h_{Q_{\pi,S}} - \sum_{\omega} h_{K_{\pi,S,\omega}} \in \mathcal{L}_{N-2}. \tag{78}$$

Substituting Equation (78) into Equation (41) expresses f as a finite linear combination of support functions of $(N-1)$-dimensional simplices in H_0, plus a remainder in $\mathcal{L}_{N-2}$. The simplex Δ contributes its support function with coefficient -1, and the linear apex terms are included in the remainder.

For a simplex P, an outward facet normal ν_F exposes the facet F on one side and its opposite vertex v_F on the other. Thus

$$D_{\nu_F} h_P(x) = h_F(x) - \langle v_F, x \rangle. \tag{79}$$

Directions exposing no facet on either side give support functions of faces of dimension at most $N-3$. Consequently, Equation (51) says that each weighted, signed sum of parallel-facet support functions belongs to $\mathcal{L}_{N-3}$.

Our assumption controls the weighted facet contributions, whereas our conclusion concerns f. To recover the original sum modulo $\mathcal{L}_{N-2}$, we use the same linear operator for every simplex, so cancellations between facet contributions survive reconstruction. Use the operator $\mathcal{T}$ defined in Equation (52). The sign in this definition concerns the input x. To interpret it geometrically, first suppose that every point $p \in P$ satisfies $p_1 \ge 0$, and write $\operatorname{pr}(p) := p - p_1 e_1$ for projection onto $p_1 = 0$. Since $\langle p, x \rangle - \langle \operatorname{pr}(p), x \rangle = p_1 x_1$, each point dominates its projection when $x_1 \ge 0$, and its projection dominates when $x_1 < 0$. Consequently,

$$\mathcal{T} h_P(x) = \max\{h_P(x), h_{\operatorname{pr} P}(x)\} = h_{\operatorname{conv}(P \cup \operatorname{pr} P)}(x). \tag{80}$$

This convex hull fills the segments with endpoints p and its projection, for $p \in P$, so its dimension is at most $\dim P + 1$. For an arbitrary P, choose $c \ge 0$ such that $P + ce_1$ lies in $p_1 \ge 0$. Translation adds cx_1 to the support function, and $\mathcal{T}(x_1) = \sigma(x_1)$, giving

$$\mathcal{T} h_P(x) = h_{\operatorname{conv}((P+ce_1) \cup \operatorname{pr}(P+ce_1))}(x) - c\sigma(x_1). \tag{81}$$

The correction $\sigma(x_1)$ is the support function of the segment $[0, e_1]$. Thus linearity gives

$$\mathcal{T} \mathcal{L}_k \subseteq \mathcal{L}_{k+1}. \tag{82}$$

Proof of Claim 10.3. Let $k = |\{F \text{ facet of } P : \langle \nu_F, e_1 \rangle \neq 0\}|$ count facets not parallel to e_1. Boundedness ensures both nonzero signs occur, so $k \ge 2$.

Start ($k = 2$). There is one upper facet F^+ and one lower facet F^-. Every line parallel to e_1 that meets P has its two intersection endpoints on these facets. Thus $h_P(x) = h_{F^+}(x)$ when $x_1 \geq 0$, and $h_P(x) = h_{F^-}(x)$ when $x_1 < 0$. When $x_1 \geq 0$, $\mathcal{T}$ evaluates at x, so subtracting $\mathcal{T}(h_{F^+} - h_{F^-})$ from h_P leaves h_{F^-}. When $x_1 < 0$, $\mathcal{T}$ evaluates both facet support functions at $x - x_1 e_1$, where they agree because the facets have the same projection. Their difference is therefore zero, again leaving h_{F^-}. The reconstruction error is therefore $h_{F^-} \in \mathcal{L}_{N-2}$, proving the claim for $k = 2$.

Step ($k > 2$). Assume the claim for $k - 1$, and let P have k facets not parallel to e_1. Two of them have the same nonzero sign of $\langle \nu_F, e_1 \rangle$. Their intersection is an $(N-3)$-dimensional face C. Let $W := \operatorname{span}\{p - q : p, q \in C\}$, and project orthogonally within H_0 onto $W^\perp$. This is a two-dimensional plane, since $\dim H_0 = N - 1$ and $\dim W = N - 3$. All points of C have the same projection c, because their differences lie in W. There are just two vertices of P outside C, so the projected simplex is a triangle with vertices c and the projections of those two vertices. The chosen facets project to its two sides meeting at c.

The chosen facet normals are orthogonal to W, so their signs against e_1 equal their signs against its projection. This projected direction is nonzero, since the chosen facets are not parallel to e_1. If both sides are upper, moving from c opposite to this direction enters the triangle; if both are lower, moving in this direction enters it. Continuing meets the interior of the opposite edge. The inverse image of the line through c in this direction is a hyperplane in H_0 containing C and parallel to e_1. It cuts the edge whose endpoints are the two vertices outside C at an interior point and divides P into two $(N-1)$-dimensional simplices. Each keeps one of the two chosen facets and replaces the other by the new facet parallel to e_1. Its other facets retain their normal directions, so each new simplex has exactly $k - 1$ facets not parallel to e_1.

Write $E(Q)$ for the reconstruction error on the left of Equation (53). Let P_+ and P_- be the two new simplices, let $R := P_+ \cap P_-$ be their common cutting facet, and write F_1, F_2 for the chosen facets and $R_F := R \cap F$ for every other facet of P. With $s_F := \operatorname{sgn}\langle \nu_F, e_1 \rangle$, applying Equation (77) to P and to every split facet gives the exact residual identity

$$E(P) = E(P_+) + E(P_-) - h_R + \mathcal{T}\left(\sum_{\substack{F \text{ facet of } P \\ F \notin \{F_1, F_2\}}} s_F h_{R_F} \right). \tag{83}$$

Indeed, the inherited facets retain their signs, while the two copies of R are parallel to e_1 and therefore have sign zero. By the induction hypothesis, $E(P_+), E(P_-) \in \mathcal{L}_{N-2}$; moreover, $\dim R = N - 2$ and $\dim R_F \leq N - 3$. Equation Equation (82) therefore puts every term on the right of Equation (83) in $\mathcal{L}_{N-2}$. This proves the claim for k and completes the induction.

□

Conclusion for f. Apply Equation (53) to every simplex support-function term in the expansion of f, keeping its original coefficient. A facet F parallel to e_1 has $\langle \nu_F, e_1 \rangle = 0$, so its term in the reconstruction formula is zero and can be omitted. For each remaining occurrence of a facet, choose ν_F outward from its simplex and put

$$u_F := \frac{\nu_F}{\langle \nu_F, e_1 \rangle}, \qquad \langle u_F, e_1 \rangle = 1. \tag{84}$$

Thus the coordinate with index 1 is normalized to 1; the coordinate with index 0 remains zero because $u_F \in H_0$. Oppositely oriented parallel normals give the same u_F. Only the normal is rescaled: the support function and its coefficient are unchanged.

Let $\mathcal{U}$ be the finite set of vectors u_F obtained this way. For each $u \in \mathcal{U}$, define G_u by adding all facet support-function terms with $u_F = u$, retaining their signs and weights. An occurrence of F as a

facet of $K_{\pi,S,\omega}$ contributes $(-1)^{N+|S|}\operatorname{sgn}\langle \nu_F, e_1\rangle\, h_F/N!$, while an occurrence as a facet of Δ contributes $-\operatorname{sgn}\langle \nu_F, e_1\rangle\, h_F$. These are precisely the terms with the same normalized normal, including all repetitions.

More explicitly, define ρ to be the sum of the linear apex terms and the weighted subdivision errors in Equation (78). Then

$$f = \frac{1}{N!}\sum_{\pi}\sum_{\varnothing\neq S\subseteq\{1,\dots,N-2\}}(-1)^{N+|S|}\sum_{\omega} h_{K_{\pi,S,\omega}} - h_\Delta + \rho, \qquad \rho\in\mathcal{L}_{N-2}. \tag{85}$$

For a simplex P, collect its facets with normalized normal u as

$$B_u(P) := \sum_{\substack{F \text{ facet of } P\\ \langle \nu_F, e_1\rangle\neq 0,\ u_F = u}} \operatorname{sgn}\langle \nu_F, e_1\rangle h_F. \tag{86}$$

An empty sum is zero. If $\langle \nu_F, e_1\rangle > 0$, the direction u_F exposes F; if it is negative, $-u_F$ exposes F. The opposite face is a single vertex, so Equation (79) gives the signed facet term up to a linear function. If neither u nor $-u$ exposes a facet of P, both exposed faces have dimension at most $N-3$. Thus $D_u h_P - B_u(P)\in\mathcal{L}_{N-3}$. Applying this to the full weighted expansion, with the same permutations and nonempty cut sets as in Equation (41), gives

$$\begin{aligned} D_u f - G_u = &\sum_{\pi,S\neq\varnothing}\frac{(-1)^{N+|S|}}{N!}\sum_{\omega}\big(D_u h_{K_{\pi,S,\omega}} - B_u(K_{\pi,S,\omega})\big)\\ &- (D_u h_\Delta - B_u(\Delta)) + D_u\rho \in \mathcal{L}_{N-3} \qquad (u\in\mathcal{U}). \end{aligned} \tag{87}$$

Together with Equation (51), relation Equation (87) puts every G_u in $\mathcal{L}_{N-3}$. By linearity, the sum of the reconstructed facet terms is $\mathcal{T}(\sum_{u\in\mathcal{U}} G_u)$, which lies in $\mathcal{L}_{N-2}$ by Equation (82). Its difference from f consists of the reconstruction and subdivision remainders, also in $\mathcal{L}_{N-2}$. Hence $f\in\mathcal{L}_{N-2}$. This proves the implication used in Step 10.2.

A.3 Coordinate deletion

Proof of Claim 10.5. We now determine which terms of the signed sum remain for $u = (0,1,\dots,1)$. Only label 0 has a different coordinate value from the others. Consequently, a tree-edge generator is nonorthogonal to u exactly when one of its endpoint labels is 0.

If label 0 is the apex label, all tree-edge generators are orthogonal to u. The maximizing face is the whole base $B_{\pi,S}$ and the minimizing face is $\{0\}$, so $D_u h_{Q_{\pi,S}} = h_{B_{\pi,S}} \in \mathcal{L}_1 \subseteq \mathcal{L}_{N-3}$. If label 0 is a tree vertex of degree at least two, at least two segment generators are nonorthogonal to u. The preceding dimension bound then gives $D_u h_{Q_{\pi,S}}\in\mathcal{L}_{N-3}$ as well. We can therefore discard both kinds of term: it suffices to retain descriptions in which label 0 is a tree vertex of degree one.

To express this condition in the permutation notation, let $p := \pi^{-1}(0)$ be the index of label 0 in the full order $(\pi(0),\dots,\pi(N-1))$. When $p\geq 1$, its tree position is $p-1$. A degree-one vertex is either an endpoint of the spine with no attached leaves or a leaf outside the spine. The retained descriptions are thus exactly

$$\begin{aligned} &p = 1, \quad 1\in S && \text{(root endpoint)},\\ &2\leq p\leq N-1, \quad p-1\notin S && \text{(leaf outside the spine)},\\ &p = N-1, \quad N-2\in S && \text{(last spine endpoint)}. \end{aligned} \tag{88}$$

Indeed, the root has only one child precisely when $1\in S$; a nonroot spine vertex has a parent and also a child unless it occupies the final tree position. Deleting label 0 from the full order now gives an order $\bar\pi$ mapping positions $\{0,\dots,N-2\}$ bijectively to labels $\{1,\dots,N-1\}$.

For $0 \le i \le N-3$, let ι_i insert a gap at the prospective pendant position $i+1$. Given a lower cut set $T \subseteq \{1,\dots,N-3\}$, the three active insertion rules are

$$\begin{aligned} \iota_i(t) &:= \begin{cases} t, & t < i+1, \\ t+1, & t \ge i+1, \end{cases} \\ \text{first end:} \quad & (p,S) = (1, \{1\} \cup \{t+1 : t \in T\}), \\ \text{pendant } i\text{:} \quad & (p,S) = (i+2, \{\iota_i(t) : t \in T\}) \qquad (0 \le i \le N-3), \\ \text{last end:} \quad & (p,S) = (N-1, T \cup \{N-2\}). \end{aligned} \tag{89}$$

In every row the upper permutation is uniquely obtained by inserting label 0 at position p in $\bar\pi$. Conversely, with t restricted to $\{1,\dots,N-3\}$, the deletion rules are $T = \{t : t+1 \in S\}$ for the first end, $T = \{t : \iota_i(t) \in S\}$ for pendant i, and $T = S \setminus \{N-2\}$ for the last end. These operations are inverse. The three cases in Equation (88) correspond respectively to the first-end row, the pendant row with $i = p-2$, and the last-end row. Thus a nonempty T has $N-2$ pendant and two endpoint preimages, whereas $T = \varnothing$ has only the two endpoints because upper cut sets are nonempty.

Extend the lower caterpillar notation to $T = \varnothing$ by setting its last spine position and every parent equal to 0, so its head begins at the root $\bar\pi(1)$. For every lower label order $\bar\pi$ put

$$\begin{aligned} C_{\bar\pi,T}(x) &:= x_{\bar\pi(0)} + U_{\bar\pi,T}(x), \\ q_T &:= \max(\{0\} \cup T), \qquad p_T(j) := \max(\{0\} \cup \{t \in T : t < j\}). \end{aligned} \tag{90}$$

For each upper preimage (π, S) of $(\bar\pi, T)$, the insertion identities and the literal factorial coefficients are

$$\begin{aligned} \text{pendant:} \quad & \frac{(-1)^{N+|S|}}{N!} = -\frac{1}{N}\frac{(-1)^{N-1+|T|}}{(N-1)!}, \quad D_u\big(x_{\pi(0)} + U_{\pi,S}\big) + C_{\bar\pi,T} \in \mathcal{L}_1, \\ \text{endpoint:} \quad & \frac{(-1)^{N+|S|}}{N!} = \frac{1}{N}\frac{(-1)^{N-1+|T|}}{(N-1)!}, \qquad D_u\big(x_{\pi(0)} + U_{\pi,S}\big) - C_{\bar\pi,T} \in \mathcal{L}_1. \end{aligned} \tag{91}$$

Thus every one of the $N-2+2 = N$ preimages contributes $1/N$ of the lower summand modulo $\mathcal{L}_{N-3}$, and together they recover its coefficient in F_{N-1}.

The extra terms are the two endpoint fibres whose deletion leaves $T = \varnothing$, hence a rooted star. By Equation (91), each contributes $1/N$ of the weighted lower summand modulo $\mathcal{L}_1$. For a fixed lower order $\bar\pi$, the nonroot positions are $J = \{0, 2, \dots, N-2\}$, including the original apex position 0. For $a \in J$, let $\bar\pi^{(a)}$ exchange positions 0 and a, keeping the root at position 1; the exchange at $a = 0$ is the identity. Put

$$\begin{aligned} q := |J| = N-2, \qquad & d_a := x_{\bar\pi(1)} - x_{\bar\pi(a)} \quad (a \in J), \\ T_A := \sigma\left(\sum_{a\in A} d_a + \sum_{a \in J\setminus A} \sigma(d_a)\right) \quad & (A \subseteq J), \\ C_{\bar\pi^{(a)},\varnothing} = x_{\bar\pi(a)} + T_{\{a\}} \quad & (a \in J). \end{aligned} \tag{92}$$

The Boolean alternating identity gives

$$\sum_{A\subseteq J} (-1)^{|A|} T_A = 0 \quad \Longrightarrow \quad \sum_{a\in J} T_{\{a\}} = T_\varnothing + \sum_{\substack{A\subseteq J \\ |A|\ge 2}} (-1)^{|A|} T_A. \tag{93}$$

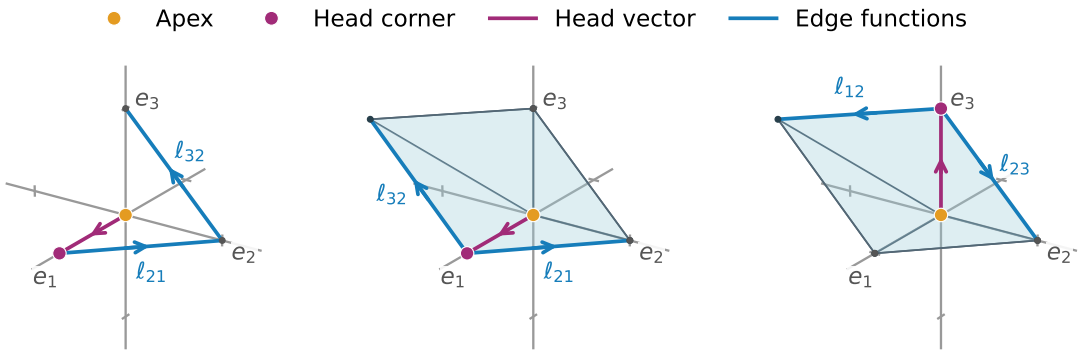
The empty term is a sum of support functions of segments. For $|A| \ge 2$, the polytope with support function T_A has one head direction and $q - |A|$ segment directions, hence dimension at most $1 + q - |A| \le q - 1 = N-3$.

Thus the singleton sum lies in $\mathcal{L}_{N-3}$; the sum of the linear apex coordinates lies in $\mathcal{L}_0$. Each fixed-position exchange is a bijection on lower label orders, so

$$q \sum_{\bar{\pi}} C_{\bar{\pi},\varnothing} = \sum_{\bar{\pi}} \sum_{a \in J} C_{\bar{\pi}^{(a)},\varnothing} \in \mathcal{L}_{q-1}. \tag{94}$$

Since $q > 0$, the full empty-cut sum belongs to $\mathcal{L}_{q-1} = \mathcal{L}_{N-3}$. The empty-cut coefficient $(-1)^{N-1}/(N-1)!$ and the factor $2/N$ from the two endpoint fibres are scalars, so the extra star sum lies there as well. This proves Claim 10.5. □

B Network construction and width bounds



(a) Coordinate differences (b) Original description (c) Normalized description

Figure 3: Coordinate differences and the caterpillar polytope for $\pi = (0, 3, 2, 1)$ and $S = \{1, 2\}$. (a) The head $h = e_1 - e_0$ and edge functions $L = \{\ell_{32} = e_3 - e_2, \ell_{21} = e_2 - e_1\}$ drawn between their labelled endpoints. (b) The same generators translated to the base corner h of $Q_{h,L}$. (c) Normalization (Section 6) gives $h^+ = h + \ell_{32} + \ell_{21} = e_3 - e_0$ and $L^+ = \{\ell_{12} = e_1 - e_2, \ell_{23} = e_2 - e_3\}$, leaving the polytope unchanged. Purple arrows represent the head (the apex–last spine connection before normalization); blue arrows represent the spine edge functions. Orange dots mark the polytope apex and purple dots the chosen head corner. The grey coordinate axes meet at the origin, representing e_0 after restriction to $x_0 = 0$; the positive unit points are labelled e_1, e_2, e_3.

Proof of Theorem 1. Fix $N \geq 3$. We bound the quotient dimension q_N of Section 7 by the count of normalized descriptions and apply Theorem 11.

Step 1.1 (Count the caterpillar face functions). Use the adjusted heads and normalized descriptions from Section 6. Counting distinct normalized descriptions bounds q_N.

For a caterpillar description (π, S), the spine endpoints are labelled $\pi(1)$ and $\pi(q+1)$, and every label other than $\pi(1)$ and the apex is the child of exactly one edge. Hence the normalized description of $U_{\pi,S}$ is

$$\begin{aligned} h^+_{\pi,S}(x) &= \sum_{\ell_{ij} \in L_{\pi,S}} x_{\max(i,j)} + x_{\pi(1)} + x_{\pi(q+1)} - \sum_{i=0}^{N-1} x_i, \\ L^+_{\pi,S} &= \{\ell_{\min(i,j),\max(i,j)} : \ell_{ij} \in L_{\pi,S}\}. \end{aligned} \tag{95}$$

This depends only on the undirected tree, apex, and unordered spine endpoints. For m spine vertices, choose the vertex set in $\binom{N}{m}$ ways, its order up to reversal in $m!/2$ ways, the apex in $N-m$ ways, and each of the $N-m-1$ leaf attachments in m ways. Reversal has no fixed point since $m \geq 2$, so Equation (6) gives

$$\left|\{(h^+_{\pi,S}, L^+_{\pi,S}) : (\pi, S) \text{ a caterpillar description}\}\right| \leq C_N. \tag{96}$$

Reversing an edge swaps its omitted and linear states. Thus choosing states before or after normalization gives the same family $\mathcal{K}_0$ of descriptions, with $|\mathcal{K}_0| \leq 3^{N-2}C_N$. Write $U_k := U_{h,M}$ for $k = (h, M)$. We merge equal descriptions; distinct descriptions may still represent the same function.

Among them are the descriptions $(\ell_{ij}, \varnothing)$ representing $x \mapsto \sigma(x_i - x_j)$, with $i \neq j$: for the all-cut path $S = \{1, \dots, r\}$ with apex $\pi(0) = j$ and last spine vertex $\pi(r+1) = i$, omitting every edge function gives

$$U^0_{\pi,S} = \sigma \circ \ell_{ij} \qquad (0 \leq i \neq j < N). \tag{97}$$

Let $\mathcal{P} \subseteq \mathcal{K}_0$ consist of descriptions with empty hinge set and head a coordinate difference. The distinct heads ℓ_{ij} give $|\mathcal{P}| \geq N(N-1)$. Put

$$K := \mathcal{K}_0 \setminus \mathcal{P}. \tag{98}$$

Every face function in $\mathcal{D}$ equals U_k for its normalized description $k \in \mathcal{K}_0$, and by Equation (8) every unit indexed by $\mathcal{P}$ lies in the auxiliary span A_N. Hence the units U_k, $k \in K$, span V_N modulo A_N, and

$$q_N \leq |K| \leq 3^{N-2}C_N - N(N-1). \tag{99}$$

Step 1.2 (Compile the network and establish its widths). Theorem 11 gives a network with rational weights, zero biases, first width h_1, and second width

$$q_N + 2 \leq 3^{N-2}C_N - N(N-1) + 2 = h_2. \tag{100}$$

Zero padding realizes the stated widths exactly.

For the growth rates, count the ordered choices of m spine labels:

$$\binom{N}{m}\frac{m!}{2} = \frac{N!}{2(N-m)!} \leq N^m \qquad (2 \leq m \leq N-1). \tag{101}$$

Using $N - m \leq N$ and $m^{N-m-1} \leq N^{N-m-1}$ in each summand of Equation (6) gives

$$\begin{aligned} C_N &\leq \sum_{m=2}^{N-1} N^m \, N \, N^{N-m-1} = (N-2)N^N \\ &\leq N^{N+1} \leq N^{4N}. \end{aligned} \tag{102}$$

Since $3^{N-2} \leq N^N$ for $N \geq 3$,

$$h_2 \leq 3^{N-2}(N-2)N^N + 2 \leq 2N^{5N}, \tag{103}$$

$$h_1 \leq 4N^2. \tag{104}$$

Taking logarithms gives the asymptotic bounds in Theorem 1. □

C Proofs of the lower bounds

C.1 Proof of the exponential count bound

Idea. Fix a tie size m with $3 \le m \le N$. A *locality* is a point at which m coordinates tie at the top value 0, while the remaining coordinates lie far below, at heights $-B^e$ for distinct exponents e. The m-th mixed finite difference in the tied coordinates kills every function that depends on at most two coordinates. Hence it kills the auxiliary span A_N, whereas it does not vanish on $\max_N$. Applying it to an expansion shows that some face function U_k, $k \in K$, is *seen* at every locality. A face function is seen only if its preactivation vanishes at the locality. Reading the preactivation in base B, this forces a rigid balance between the heights of the coordinates and the functions $\sigma(x_i - x_j)$ inside U_k. These functions lie on a caterpillar, so the balance leaves few tie sets and few orders. Counting shows that each face function is seen at no more than a fraction $6^m/\binom{N}{m}$ of the localities.

Notation. Let $k = (h, M) \in \mathcal{K}_0$ be a normalized description, so that $U_k = \sigma(A_k)$ with preactivation $A_k(x) = h(x) + \sum_{\ell_{ij} \in M} \sigma(x_i - x_j)$, where $i < j$ for every $\ell_{ij} \in M$ and $h(x) = \sum_o h_o x_o$ has integer coefficients. The *hinge graph* of k has the labels $0, \dots, N-1$ as vertices and an edge $\{i, j\}$ for every $\ell_{ij} \in M$; vertices joined by an edge are *hinge neighbours*. For every label o put $\kappa_o := h_o + \big|\{j : \ell_{oj} \in M\}\big|$. This is the slope of A_k in the direction x_o at points where x_o exceeds the coordinates of all hinge neighbours of o. By Equation (8), the normalization in Equation (21) does not change A_k, so κ_o equals the coefficient of x_o in the head h' of Equation (20) plus the number of hinges $\sigma(x_o - x_j)$ in M before reorientation.

Encode a locality by a permutation ρ of $\{0, \dots, N-1\}$, read as an assignment of *positions*: label o sits at position $\rho(o)$, and a smaller position means a higher coordinate. The *tie set* is $T_\rho := \{o : \rho(o) < m\}$, so $|T_\rho| = m$. For integers $B \ge 1$ and a real $\delta > 0$, chosen below, define the base point $p^\rho \in \mathbb{R}^N$ and the mixed difference Δ_ρ by

$$p^\rho_o := \begin{cases} 0, & \rho(o) < m, \\ -B^{\rho(o)}, & \rho(o) \ge m, \end{cases} \tag{105}$$

$$\Delta_\rho g := \sum_{S \subseteq T_\rho} (-1)^{|T_\rho \setminus S|}\, g\big(p^\rho + \delta \mathbf{1}_S\big), \tag{106}$$

where $\mathbf{1}_S$ is the indicator vector of S. Thus Δ_ρ evaluates g at the 2^m corners of a cube of side δ in the tied coordinates. Each locality $(T, \text{order of the other labels})$ arises from exactly $m!$ permutations ρ, so we count permutations. For $o \notin T_\rho$ let $u_\rho(o)$ be the number of hinge neighbours of o at a smaller position. Say that k *passes* ρ if

$$u_\rho(o) = \kappa_o \qquad \text{for every } o \notin T_\rho, \tag{107}$$

$$\kappa_t \ne 0 \qquad \text{for every } t \in T_\rho \text{ without hinge neighbours in } T_\rho. \tag{108}$$

Proof of Theorem 8. Fix an expansion $\max_N = \sum_{k \in K} d_k U_k + R$ as in Equation (23), and a tie size $3 \le m \le N$. Put $c_k := \sum_o |h_o| + 2|M|$ for $k = (h, M)$, $C := \sum_{k \in K} c_k$, $\delta := 1/(C+1)$, and $B := \lceil C \rceil + 1$. Then $\delta c_k < 1$ and $\sum_o |h_o| + |M| < B$ for every $k \in K$. Steps 8.1 to 8.4 show that every ρ is passed by some $k \in K$. Steps 8.5 to 8.9 show that each k passes at most $m!\,(N-m)!\,6^m$ permutations.

Step 8.1 (The mixed difference kills the auxiliary span). If g does not depend on some tied coordinate x_t on the cube, then $\Delta_\rho g = 0$: the corners S and $S \cup \{t\}$, $t \notin S$, carry the same value with opposite signs. Since $m \ge 3$, every function of at most two coordinates misses some tied coordinate. This applies to the auxiliary

functions Equation (18), to the coordinates x_o, and to every $\sigma(x_i - x_j)$. On the cube the coordinates outside T_ρ are at most -1, so $\max_N(p^\rho + \delta \mathbf{1}_S) = \delta$ for $S \neq \varnothing$ and 0 for $S = \varnothing$. As the signs sum to zero,

$$\Delta_\rho R = 0 \qquad (R \in A_N), \tag{109}$$

$$\Delta_\rho \max_N = -(-1)^m \delta. \tag{110}$$

Step 8.2 (A face function is seen only where its preactivation vanishes). Let $k = (h, M) \in K$. The base point has integer coordinates, so $A_k(p^\rho)$ is an integer. On the cube every coordinate moves by at most δ, so h changes by at most $\delta \sum_o |h_o|$ and each $\sigma(x_i - x_j)$ by at most 2δ. Hence A_k stays within $\delta c_k < 1$ of $A_k(p^\rho)$. If $A_k(p^\rho) \leq -1$, then $U_k = 0$ on the cube. If $A_k(p^\rho) \geq 1$, then $U_k = A_k$ on the cube, and A_k is a sum of functions of at most two coordinates. In both cases $\Delta_\rho U_k = 0$ by Step 8.1. Now let $A_k(p^\rho) = 0$, and let $t \in T_\rho$ have no hinge neighbour in T_ρ. Raising x_t from 0 to δ activates no new function: every $\sigma(x_t - x_o)$ in M has $x_o \leq -1$ and stays active with slope 1, and every $\sigma(x_o - x_t)$ stays 0. So A_k changes by $\delta\kappa_t$. If $\kappa_t = 0$, then U_k does not depend on x_t on the cube and $\Delta_\rho U_k = 0$. Hence $\Delta_\rho U_k \neq 0$ implies $A_k(p^\rho) = 0$ and Equation (108).

Step 8.3 (A vanishing preactivation balances the heights). At p^ρ, the function $\sigma(x_i - x_j)$ equals $p_i^\rho - p_j^\rho$ when i is at a smaller position than j, and 0 otherwise. Collecting the coefficient of each p_o^ρ and using $p_t^\rho = 0$ on T_ρ gives

$$A_k(p^\rho) = -\sum_{o \notin T_\rho} \big(\kappa_o - u_\rho(o)\big) B^{\rho(o)}. \tag{111}$$

The integer digits satisfy $|\kappa_o - u_\rho(o)| \leq |h_o| + |M| < B$, and the exponents $\rho(o)$ are distinct. If the sum vanishes, the digit with the smallest exponent is divisible by B, hence zero, and induction makes every digit zero. Thus $A_k(p^\rho) = 0$ implies Equation (107).

Step 8.4 (Every locality is passed). Apply Δ_ρ to the expansion. The remainder contributes nothing by Equation (109), and the maximum contributes $-(-1)^m\delta$ by Equation (110). Hence $\sum_{k \in K} d_k \Delta_\rho U_k \neq 0$, and some $k \in K$ has $\Delta_\rho U_k \neq 0$. By Steps 8.2 and 8.3, this k passes ρ.

Step 8.5 (The hinge graph lies on a caterpillar). Let k be a normalized description of the face function $U^\tau_{\pi,S}$. Every $\ell_{ij} \in M$ comes from an edge of the labelled caterpillar of Definition 3 with state $\sigma \circ \ell$. Let $s_0, \ldots, s_n$, $n := |S|$, be the labels at the spine positions in order. The hinge graph has two kinds of edges: *links* $\{s_i, s_{i+1}\}$, for spine edges kept as hinges, and *pendant edges* $\{s_i, u\}$, for leaves u whose parent edge to s_i is kept as a hinge. A pendant vertex u has no other hinge neighbour, and $\kappa_u = 0$: the head h' of Equation (20) does not involve u, since u is neither the apex nor the last spine label and its only edge is a hinge, and u is the subtracted end of its hinge $\sigma(x_{s_i} - x_u)$. Every other label is *isolated*, that is, without hinge neighbours.

Step 8.6 (Passed tie sets are determined by their spine part). Let k pass ρ and put $T := T_\rho$. A pendant u at s_i lies in T exactly when s_i does. If $u \in T$ and $s_i \notin T$, then u has no hinge neighbour in T and $\kappa_u = 0$, contradicting Equation (108). If $u \notin T$ and $s_i \in T$, then $u_\rho(u) = 1 \neq 0 = \kappa_u$, contradicting Equation (107). An isolated label o lies in T exactly when $\kappa_o \neq 0$, by Equation (108) for $o \in T$ and by Equation (107), which reads $0 = \kappa_o$, for $o \notin T$. Hence T is determined by its *spine part* $A_T := \{i : s_i \in T\}$, and $|A_T| \leq |T| = m$. Moreover A_T lies in the family $\mathcal{A}$ of sets $A \subseteq \{0, \ldots, n\}$ with $i \in A \Leftrightarrow \kappa_{s_i} \neq 0$ for every isolated spine label s_i.

Step 8.7 (Few orders balance a given tie set). Fix T as in Step 8.6. Among the permutations ρ with $T_\rho = T$, consider those satisfying Equation (107). For $o \notin T$, all hinge neighbours of o in T lie above o, so Equation (107) prescribes how many hinge neighbours of o outside T lie above o. The hinge edges outside T form a forest, by Step 8.5. Orient each of them towards its higher end. Then the out-degrees of

this orientation are prescribed, and a forest orientation is determined by its out-degrees: the edge at a leaf is oriented by the leaf's out-degree, and one removes the leaf and repeats. Hence all these ρ orient each hinge edge outside T in the same way. Now choose disjoint hinge edges outside T greedily: link i is chosen when $i, i+1 \notin A_T$ and link $i-1$ was not chosen, and a pendant edge is chosen at every s_i, $i \notin A_T$, that has a pendant but no link. Its pendant lies outside T by Step 8.6. Let $\mu(A_T)$ be the number of chosen edges. Swapping the two ends of one chosen edge preserves T_ρ, so exactly a $2^{-\mu(A_T)}$ fraction of the $m!\,(N-m)!$ permutations with tie set T orients all chosen edges in a prescribed way. Therefore

$$\big|\{\rho : T_\rho = T,\ \rho \text{ satisfies Equation (107)}\}\big| \cdot 2^{\mu(A_T)} \le m!\,(N-m)!. \tag{112}$$

Step 8.8 (The generating function along the spine). For $0 \le j \le n$ and $A \subseteq \{0, \dots, j\}$, let $\mu_j(A)$ count the links chosen among $0, \dots, j-1$ and the pendant edges chosen at $s_0, \dots, s_j$. Call s_j *free* if $j \notin A$ and link $j-1$ was not chosen, and define the potential

$$\phi_j(A) := \begin{cases} 7/9, & \text{link } j \text{ exists and } s_j \text{ is free,} \\ 7/6, & \text{link } j \text{ exists and } s_j \text{ is not free,} \\ 1, & \text{there is no link } j. \end{cases} \tag{113}$$

Let $P_j := \sum_A 2^{-\mu_j(A)} 6^{-|A|} \phi_j(A)$ over the sets $A \subseteq \{0, \dots, j\}$ obeying the rule of $\mathcal{A}$ for isolated s_i, $i \le j$. Then $\mu_n = \mu$. We show $P_j \le 1$ by induction on j. Write the term of A as $\psi\,\phi_j(A)$ with $\psi := 2^{-\mu_j(A)} 6^{-|A|}$. Each set for $j+1$ arises from a set A for j by omitting or inserting $j+1$, and it suffices that the allowed extensions of A have terms summing to at most $\psi\,\phi_j(A)$. Inserting $j+1$ multiplies ψ by $1/6$ and leaves s_{j+1} not free, so it contributes $\psi\lambda/6$ with $\lambda = 7/6$ if link $j+1$ exists and $\lambda = 1$ otherwise. If link j exists and s_j is free, omitting $j+1$ chooses link j, which contributes $\psi\lambda/2$, and $\lambda/2 + \lambda/6 \le 7/9$. If link j exists and s_j is not free, the two terms are at most ψ times $7/9 + 7/36$ or $1 + 1/6$, both at most $7/6$. If there is no link j, the two terms are at most ψ times $7/9 + 7/36$ if link $j+1$ exists, $1/2 + 1/6$ if s_{j+1} has a pendant but no link, and $\max(1, 1/6)$ if s_{j+1} is isolated, because then only one extension is allowed. The case $j = 0$ is the last case started from the empty set. Since no link leaves s_n, $\phi_n = 1$, and $P_n \le 1$ reads

$$\sum_{A \in \mathcal{A}} 2^{-\mu(A)}\, 6^{-|A|} \le 1. \tag{114}$$

Step 8.9 (Each face function passes few permutations). By Step 8.6, the permutations passed by k have admissible tie sets of size m. Each passed ρ satisfies Equation (107), so it is counted in Equation (112) for $T = T_\rho$. Summing Equation (112) over these tie sets, the number of permutations passed by k is at most $m!\,(N-m)! \sum_T 2^{-\mu(A_T)}$. Distinct tie sets have distinct spine parts in $\mathcal{A}$, and $|A_T| \le m$ gives $1 \le 6^{m-|A_T|}$. Hence the sum is at most $6^m \sum_{A \in \mathcal{A}} 2^{-\mu(A)} 6^{-|A|}$, which is at most 6^m by Equation (114). So k passes at most $m!\,(N-m)!\,6^m$ permutations.

Step 8.10 (Count all localities). By Step 8.4, the $N!$ permutations are covered by the sets of permutations passed by the $k \in K$. By Step 8.9, $N! \le |K|\, m!\,(N-m)!\,6^m$, that is, $\binom{N}{m} \le 6^m |K|$. This is Equation (31).

Step 8.11 (Sum over all tie sizes). For $m \le 2$ the quadratic bound Equation (33) gives $\binom{N}{m} \le 6^m |K|$, using $N(N-1) \ge 6$ for $m = 0$. The quadratic bound is proved like Step 8.10, with three-element tie sets, a six-point second difference in two of the tied coordinates, and the finer count of at most $N-2$ tie sets per face function. Together with Equation (31), the binomial theorem gives

$$7^N = \sum_{m=0}^{N} \binom{N}{m} 6^{N-m} \le \sum_{m=0}^{N} 6^{N-m}\, 6^m |K| = (N+1)\, 6^N |K|. \tag{115}$$

This is Equation (32). □

C.2 Proof of the dimension bound

Idea. For each order of the $N-2$ middle labels we take one caterpillar function whose caterpillar is a path. A second difference in the direction x_0 detects the kink of a face function along a line. We place two lines, a test line and a reference line, so that every auxiliary function has the same kinks on both. The difference of the two second differences then kills A_N, and on the test line of one path only the unit of that path has a kink.

Proof of Theorem 9. Put $b := N-1$ and $\varepsilon := 1/(2(N-1))$.

Step 9.1 (Path units). Let π be a permutation of $\{0,\dots,N-1\}$ fixing 0 and b; there are $(N-2)!$ of them. The caterpillar description $(\pi, \{1,\dots,N-2\})$ cuts at every position, so its caterpillar is the path $\pi(1),\dots,\pi(N-1) = b$ with apex 0. Keeping all edges as hinges gives the face function $U_\pi := U_{\pi,\{1,\dots,N-2\}} \in V_N$ with

$$U_\pi(x) = \sigma\Big(x_b - x_0 + \sum_{j=1}^{N-2} \sigma\big(x_{\pi(j)} - x_{\pi(j+1)}\big)\Big). \tag{116}$$

Step 9.2 (A functional that kills the auxiliary span). For a permutation ρ fixing 0 and b and for $t \in \mathbb{R}$, define the *test configuration* $x^\rho(t)$ and the *reference configuration* $y(t)$ by $x^\rho_0(t) = y_0(t) = 1+t$ and

$$x^\rho_{\rho(j)}(t) = \frac{j}{N-1} \qquad (1 \le j \le N-1), \tag{117}$$

$$y_b(t) = 1, \qquad y_i(t) = 2 \qquad (1 \le i \le N-2). \tag{118}$$

Only x_0 moves, and in both configurations $x_b = 1$ while every other coordinate except x_0 differs from 1 by at least 2ε. Define the symmetric second difference J and the functional Λ_ρ by $Jg(z) := g(z(\varepsilon)) + g(z(-\varepsilon)) - 2g(z(0))$ and $\Lambda_\rho g := Jg(x^\rho) - Jg(y)$. The second difference vanishes on functions that are affine in t on $[-\varepsilon,\varepsilon]$, and $J\sigma(a-t) = \varepsilon$ if $a = 0$. Among the auxiliary functions Equation (18), only $\sigma(x_0 - x_b)$ has a kink in $[-\varepsilon,\varepsilon]$, in both configurations alike: $\sigma(\pm x_0)$ is affine since $|t| < 1$, functions not involving x_0 are constant, and $\sigma(x_0 - x_j) = \sigma(1 + t - z_j)$ kinks at $t = z_j - 1$. Hence $\Lambda_\rho R = 0$ for every $R \in A_N$.

Step 9.3 (The functional on path units). Let z be either configuration, and let $\Sigma_\pi(z) \ge 0$ be the inner sum in Equation (116), which does not involve x_0. Since $z_b = 1$, $U_\pi(z(t)) = \sigma(\Sigma_\pi(z) - t)$. Its second difference is ε if $\Sigma_\pi(z) = 0$ and 0 if $\Sigma_\pi(z) \ge \varepsilon$. At y the last inner term is $\sigma(2-1) = 1$, so $JU_\pi(y) = 0$. At x^ρ the inner sum vanishes when the values increase along the path of π, that is, when $\pi = \rho$. Otherwise $\rho^{-1} \circ \pi$ is not the identity and has a descent at two consecutive path positions, which contributes at least $1/(N-1) = 2\varepsilon$, so $JU_\pi(x^\rho) = 0$. Therefore

$$\Lambda_\rho U_\pi = \begin{cases} \varepsilon, & \pi = \rho, \\ 0, & \pi \ne \rho. \end{cases} \tag{119}$$

Step 9.4 (Independence modulo the auxiliary span). If $\sum_\pi c_\pi U_\pi \in A_N$, then applying Λ_ρ and Steps 9.2 and 9.3 gives $c_\rho \varepsilon = 0$ for every ρ. Thus the $(N-2)!$ path units are linearly independent modulo A_N, and $q_N = \dim(V_N/A_N) \ge (N-2)!$. □

D Relation to the construction of Brunck et al.

The combinatorial proof of [BHR+26, Part 2] cones parallelepipeds $B_{v,\pi,M}$ to pyramids and expresses the simplex support function as a signed sum of their support functions modulo lower-dimensional terms

[BHR+26, Lemma 22]. This appendix identifies these building blocks. For a nonempty set M of marked positions, the pyramid has the support function of a translated caterpillar polytope, with apex v, label order π, and cut set $M-1$. For $M=\varnothing$ it gives a rooted star, which is not a caterpillar function. We use the indexing of [BHR+26, Section 7.1], with their n written as N and $[N]$ identified with our label set $\{0,\dots,N-1\}$. Positions of an ordering are numbered from 1.

Definition 13 (Parallelepipeds of Brunck et al., `Lean_28`, `Lean_29`)**.** Let $N\ge 3$, let v be a label, let $\pi=(\pi_1,\dots,\pi_{N-1})$ be an ordering of the remaining labels, and let $M\subseteq\{2,\dots,N-1\}$. For $p\in\mathbb{R}^N$ put $u_i(p):=\sum_{1\le j<i}p_{\pi_j}$. Following [BHR+26, Equation (3)], define

$$H_{-v}:=\Big\{p\in\mathbb{R}^N : p_v=0,\ \sum_k p_k=1\Big\}, \tag{120}$$

$$B_{v,\pi,M}:=\left\{p\in H_{-v}:\ \begin{array}{ll} 0\le u_i(p)\le 1 & \text{for } i\in M,\\ -1\le p_{\pi_i}\le 0 & \text{for } i\in\{2,\dots,N-1\}\setminus M\end{array}\right\}. \tag{121}$$

The pyramid with apex e_v is $\mathrm{pyr}_{e_v}(B_{v,\pi,M}):=\mathrm{conv}(\{e_v\}\cup B_{v,\pi,M})$, with support function $h_{v,\pi,M}$ as in Equation (16). Its last marked position and the marked parent of a position i are

$$q:=\max(\{1\}\cup M),\qquad r(i):=\max\big(\{1\}\cup\{j\in M: j<i\}\big)\qquad (2\le i\le N-1). \tag{122}$$

For $M\neq\varnothing$, $q=\max M$. For labels $a\neq c$, the *rooted star* with apex a and root c is

$$S_{a,c}(x):=\sigma\Big(x_c-x_a+\sum_{l\notin\{a,c\}}\sigma(x_c-x_l)\Big). \tag{123}$$

Theorem 14 (Relation to Brunck et al., `Lean_30`, `Lean_31`, `Lean_32`, `Lean_33`, `Lean_34`, `Lean_35`, `Lean_36`, `Lean_37`, Proof)**.** *Let v, π, and M be as in Definition 13, and let q and $r(i)$ be as in Equation* (122)*.*

(a) The parallelepiped is the zonotope

$$B_{v,\pi,M}=e_{\pi_q}+\sum_{i=2}^{N-1}\big[0,e_{\pi_{r(i)}}-e_{\pi_i}\big]. \tag{124}$$

(b) For every $x\in\mathbb{R}^N$,

$$h_{v,\pi,M}(x)=x_v+\sigma\Big(x_{\pi_q}-x_v+\sum_{i=2}^{N-1}\sigma\big(x_{\pi_{r(i)}}-x_{\pi_i}\big)\Big). \tag{125}$$

(c) Let $M\neq\varnothing$. Define the caterpillar description (π',S) by $\pi'(0):=v$, $\pi'(j):=\pi_j$ for $1\le j\le N-1$, and $S:=\{i-1: i\in M\}$. Then $|S|=|M|$ and

$$h_{v,\pi,M}(x)=x_v+U_{\pi',S}(x)\qquad (x\in\mathbb{R}^N). \tag{126}$$

Hence $h_{v,\pi,M}-x_v=U_k$ for some $k\in\mathcal{K}_0$. Conversely, every caterpillar description (π',S) arises in this way from $v=\pi'(0)$, $\pi_j=\pi'(j)$, and $M=\{j+1: j\in S\}\neq\varnothing$.

(d) Let $M=\varnothing$. Then $h_{v,\pi,\varnothing}(x)=x_v+S_{v,\pi_1}(x)$. No rooted star is a caterpillar function: $S_{a,c}\neq U_k$ for all labels $a\neq c$ and all $k\in\mathcal{K}_0$. In particular, $h_{v,\pi,\varnothing}-x_v\neq U_k$ for every $k\in\mathcal{K}_0$.

By Equation (17), the right-hand side of Equation (126) is the support function of the translated caterpillar polytope $e_v + Q_{h_{\pi',S}, L_{\pi',S}}$. The family of [BHR+26] therefore consists of the caterpillar polytopes, translated by e_v, and of the pyramids over rooted stars. The sign $(-1)^{|M|}$ of [BHR+26, Lemma 22] equals $(-1)^{|S|}$. The theorem compares building blocks only. It makes no statement about corrections or network coefficients.

Proof of Theorem 14. Write $I := \{2, \dots, N-1\}$ and $g_i := e_{\pi_{r(i)}} - e_{\pi_i}$ for $i \in I$. Every element of $\{1\} \cup M$ lies in $\{1, \dots, N-1\}$, so q and all $r(i)$ are positions. Since $M \subseteq I$, the definitions give

- $1 \le r(i) < i$ for $i \in I$, and $i \le r(l)$ whenever $i \in M$ and $i < l$,
- $i \le q$ for $i \in M$,
- $q \ne i$ and $r(l) \ne i$ for $i \in I \setminus M$ and all l.

Step 14.1 (Coordinates of the zonotope). For $t \in [0,1]^I$ let $z(t) := e_{\pi_q} + \sum_{l \in I} t_l g_l$. Every e_{π_j} has coordinate sum 1 and vanishes at v, so $z(t) \in H_{-v}$. For $i \in I \setminus M$ the coordinate $z(t)_{\pi_i}$ collects the term $-t_i$ and no other term, since π_i is neither π_q nor any $\pi_{r(l)}$. Hence $z(t)_{\pi_i} = -t_i$. For $i \in M$, the prefix sum u_i counts a vector e_{π_j} exactly when $j < i$. Thus e_{π_q} contributes 0, because $q \ge i$. A generator g_l with $l < i$ contributes $1 - 1 = 0$, because $r(l) < l < i$. A generator with $l > i$ contributes $0 - 0$, because $r(l) \ge i$. Finally g_i contributes t_i, because $r(i) < i$. Hence $u_i(z(t)) = t_i$. Therefore $z(t) \in B_{v,\pi,M}$.

Step 14.2 (The coordinates determine the point). Let $p, p' \in H_{-v}$ with $u_i(p) = u_i(p')$ for $i \in M$ and $p_{\pi_i} = p'_{\pi_i}$ for $i \in I \setminus M$. Put $d := p - p'$ and $U_i := \sum_{1 \le j < i} d_{\pi_j}$. Then $U_1 = 0$, and $U_N = \sum_k d_k - d_v = 0$. Descending from $i = N-1$ to $i = 2$, either $i \in M$ and $U_i = 0$ by assumption, or $i \notin M$ and $U_i = U_{i+1} - d_{\pi_i} = U_{i+1} = 0$. Hence $d_{\pi_j} = U_{j+1} - U_j = 0$ for $1 \le j \le N-1$, and $d_v = 0$. Thus $p = p'$. This is the invertibility in [BHR+26, Claim 21].

Step 14.3 (Proof of (a)). Step 14.1 shows that the right-hand side of Equation (124) lies in $B_{v,\pi,M}$. Conversely, let $p \in B_{v,\pi,M}$, and set $t_i := u_i(p)$ for $i \in M$ and $t_i := -p_{\pi_i}$ for $i \in I \setminus M$. The inequalities in Equation (121) say $t \in [0,1]^I$. By Step 14.1, the points p and $z(t)$ of H_{-v} have the same coordinates, so $p = z(t)$ by Step 14.2.

Step 14.4 (Proof of (b)). A linear function attains the same supremum on a set and on its convex hull. Hence $h_{v,\pi,M}(x) = \max(x_v, \sup_{p \in B_{v,\pi,M}} \langle p, x \rangle)$. Put $b_l := x_{\pi_{r(l)}} - x_{\pi_l}$. By (a), $\langle z(t), x \rangle = x_{\pi_q} + \sum_l t_l b_l$, and $t_l b_l \le \sigma(b_l)$ for $0 \le t_l \le 1$, with equality for $t_l = \mathbb{1}_{\{b_l \ge 0\}}$. So the supremum over $B_{v,\pi,M}$ equals $A := x_{\pi_q} + \sum_l \sigma(b_l)$. Then Equation (7) gives $\max(x_v, A) = x_v + \sigma(A - x_v)$, which is Equation (125).

Step 14.5 (Proof of (c)). With π' and S as in (c), position j of the caterpillar carries the label $\pi'(j+1) = \pi_{j+1}$, so caterpillar position j is the position $j+1$ of [BHR+26]. The map $i \mapsto i - 1$ is a bijection from M onto S, hence $|S| = |M|$, and it carries $\{1\} \cup M$ onto $\{0\} \cup S$ and $\{1\} \cup \{j \in M : j < i\}$ onto $\{0\} \cup \{j \in S : j < i - 1\}$. Comparing with Equation (10), the last spine position is $q - 1$ and the parent of caterpillar position $i - 1$ is $r(i) - 1$. By Equations (12) and (13), the head is $x_{\pi_q} - x_v$ and the edge functions are $x_{\pi_{r(i)}} - x_{\pi_i}$, $i \in I$. So the right-hand side of Equation (125) is $x_v + U_{\pi',S}(x)$. If $M \ne \varnothing$, then $S \ne \varnothing$ and $0 \notin S$, so (π', S) is a caterpillar description, and its normalized description $k \in \mathcal{K}_0$ satisfies $U_k = U_{\pi',S}$ by Equation (8). The converse inverts these bijections.

Step 14.6 (Proof of (d)). For $M = \varnothing$, $q = 1$ and $r \equiv 1$, and $\{\pi_i : i \in I\}$ is the set of labels other than v and π_1. So Equation (125) becomes $x_v + S_{v,\pi_1}(x)$. Let $a \ne c$ and evaluate at $x = e_c$. Every inner term of $S_{a,c}(e_c)$ equals $\sigma(1)$, so $S_{a,c}(e_c) = 1 + (N-2) = N - 1$. Let $k \in \mathcal{K}_0$. Then U_k is the normalization of a face function $U^{\tau}_{\pi',S}$, and $U_k = U^{\tau}_{\pi',S}$ by Equation (8). Since $\ell \le \sigma \circ \ell$, $0 \le \sigma \circ \ell$, and σ is monotone, $U^{\tau}_{\pi',S} \le U_{\pi',S}$. At $x = e_c$, every head and edge function takes values in $\{-1, 0, 1\}$. If $c = \pi'(0)$, the argument of the outer σ in $U_{\pi',S}(e_c)$ is -1, so $U_{\pi',S}(e_c) = 0$. Otherwise c is the label of a caterpillar position

m. The head contributes at most 1, and every inner hinge at most 1. If $m \geq 1$, the inner hinge of the edge into position m is $\sigma(x_{\pi'(p(m)+1)} - 1) = 0$, so the value is at most $1 + (N - 3)$. If $m = 0$, the head contributes 0, because $q \geq \min S > 0$, so the value is at most $N - 2$. Thus $U_k(e_c) \leq N - 2 < S_{a,c}(e_c)$. □

E Lean definitions and main theorem

The real parameter arrays in Equation (4) have dimensions

$$\begin{aligned} W_1 &\in \mathbb{R}^{h_1 \times N}, \quad b_1 \in \mathbb{R}^{h_1}, \\ W_2 &\in \mathbb{R}^{h_2 \times h_1}, \quad b_2 \in \mathbb{R}^{h_2}, \\ w_3 &\in \mathbb{R}^{h_2}, \qquad b_3 \in \mathbb{R}, \end{aligned} \tag{127}$$

Sources (`Definitions.lean`, `Theorems.lean`). The listings specify the network, widths, and correctness for every real input. They reproduce declarations from these files at revision `8b4d74ab66f4`, omitting imports, documentation comments, and the theorem's proof body; these remain in the linked sources.

E.1 Network and width definitions

```
namespace MaxCaterpillar

open BigOperators

abbrev Point (N : ℕ) := Fin N → ℝ

noncomputable def relu (t : ℝ) : ℝ := max 0 t

structure Network (N h₁ h₂ : ℕ) where
  W₁ : Fin h₁ → Fin N → ℝ
  b₁ : Fin h₁ → ℝ
  W₂ : Fin h₂ → Fin h₁ → ℝ
  b₂ : Fin h₂ → ℝ
  w₃ : Fin h₂ → ℝ
  b₃ : ℝ

noncomputable def Network.eval {N h₁ h₂ : ℕ} (net : Network N h₁ h₂)
    (x : Point N) : ℝ :=
  (∑ k, net.w₃ k * relu
    ((∑ j, net.W₂ k j * relu
      ((∑ i, net.W₁ j i * x i) + net.b₁ j)) + net.b₂ k))
    + net.b₃

def Network.HasRationalWeights {N h₁ h₂ : ℕ}
    (net : Network N h₁ h₂) : Prop :=
  (∀ j i, ∃ q : ℚ, (q : ℝ) = net.W₁ j i) ∧
    (∀ k j, ∃ q : ℚ, (q : ℝ) = net.W₂ k j) ∧
    ∀ k, ∃ q : ℚ, (q : ℝ) = net.w₃ k
```

```
def Network.HasZeroBiases {N h₁ h₂ : ℕ}
    (net : Network N h₁ h₂) : Prop :=
  net.b₁ = 0 ∧ net.b₂ = 0 ∧ net.b₃ = 0

noncomputable def Maximum (N : ℕ) [NeZero N] (x : Point N) : ℝ :=
  Finset.univ.sup'
    ⟨⟨0, Nat.pos_of_ne_zero (NeZero.ne N)⟩,
      Finset.mem_univ _⟩ x

def firstWidth (N : ℕ) : ℕ := N.choose 2 + N + 1

def coneCount (N : ℕ) : ℕ :=
  ∑ m in Finset.Icc 2 (N - 1),
    N.choose m * (m.factorial / 2) * (N - m)
      * m ^ (N - m - 1)

def secondWidth (N : ℕ) : ℕ :=
  coneCount N * 3 ^ (N - 2) - N * (N - 1) + 2

end MaxCaterpillar
```

E.2 Caterpillar functions and expansions

These definitions, also in `Definitions.lean`, state the results of Section 4. A description $k = (h, M)$ is stored in reduced coordinates, and `realUnitY` evaluates U_k on $\mathbb{R}^N$.

```
abbrev FaceKey (r : ℕ) :=
  MaxNN.Caterpillar.Corrections.RealKey (r + 1)

noncomputable def caterpillarDescriptions (r : ℕ) :
    Finset (FaceKey r) :=
  MaxNN.Caterpillar.Corrections.normalizedKeys r

noncomputable def caterpillarFunction {r : ℕ}
    (k : FaceKey r) : Point (r + 1 + 1) → ℝ :=
  MaxNN.Caterpillar.Corrections.realUnitY k

noncomputable def auxiliaryFunction (N : ℕ)
    (j : Fin (MaxNN.Width.W N)) : Point N → ℝ :=
  fun x => MaxNN.Width.feat N x j

def IsCaterpillarExpansion (r : ℕ)
    (K : Finset (FaceKey r)) : Prop :=
  K ⊆ caterpillarDescriptions r ∧
    ∃ (d : FaceKey r → ℝ)
      (a : Fin (MaxNN.Width.W (r + 1 + 1)) → ℝ),
      ∀ x, Maximum (r + 1 + 1) x =
        ∑ k in K, d k * caterpillarFunction k x +
```

```
      ∑ j, a j * auxiliaryFunction (r + 1 + 1) j x

noncomputable def auxiliarySpan (N : ℕ) :
    Submodule ℝ (Point N → ℝ) :=
  Submodule.span ℝ (Set.range (auxiliaryFunction N))

noncomputable def caterpillarSpan (r : ℕ) :
    Submodule ℝ (Point (r + 1 + 1) → ℝ) :=
  Submodule.span ℝ
    ((caterpillarFunction '' ↑(caterpillarDescriptions r))
      ∪ Set.range (auxiliaryFunction (r + 1 + 1)))

noncomputable def caterpillarDim (r : ℕ) : ℕ :=
  FiniteDimensional.finrank ℝ (↥(caterpillarSpan r) /
    (auxiliarySpan (r + 1 + 1)).comap
      (caterpillarSpan r).subtype)
```

E.3 The main theorems

The assumption `[NeZero N]` expresses positive arity. For $N \geq 3$ the first statement is Theorem 1; the natural-number width definitions also cover $N = 1, 2$. With $N = r + 2$, the next three are Theorem 8, and the last is Theorem 9. All are in `Theorems.lean`.

```
theorem caterpillar_net (N : ℕ) [NeZero N] :
    ∃ net : Network N (firstWidth N) (secondWidth N),
      (∀ x, net.eval x = Maximum N x) ∧
        net.HasRationalWeights ∧ net.HasZeroBiases

theorem caterpillar_expansion_choose_le (r : ℕ)
    {K : Finset (FaceKey r)}
    (hK : IsCaterpillarExpansion r K)
    {k : ℕ} (hk : 3 ≤ k) (hkN : k ≤ r + 1 + 1) :
    (r + 1 + 1).choose k ≤ 6 ^ k * K.card

theorem caterpillar_expansion_exponential (r : ℕ)
    (hr : 1 ≤ r) {K : Finset (FaceKey r)}
    (hK : IsCaterpillarExpansion r K) :
    7 ^ (r + 1 + 1) ≤
      (r + 1 + 1 + 1) * 6 ^ (r + 1 + 1) * K.card

theorem caterpillar_expansion_quadratic (r : ℕ)
    (hr : 1 ≤ r) {K : Finset (FaceKey r)}
    (hK : IsCaterpillarExpansion r K) :
    (r + 1 + 1) * (r + 1) ≤ 6 * K.card

theorem caterpillarDim_ge (r : ℕ) (hr : 1 ≤ r) :
    r.factorial ≤ caterpillarDim r
```

E.4 The building blocks of Brunck et al.

The definitions of Definition 13, in the indexing of [BHR+26, Section 7.1], are in `Competing/Defs.lean`. An ordering is a bijection from the $N-1$ positions onto the labels other than v, and `Ordering.pos` returns π_i at position i, $1 \le i \le N-1$.

```
noncomputable def supportFunction {n : ℕ}
    (P : Set (Fin n → ℝ)) (y : Fin n → ℝ) : ℝ :=
  sSup ((fun p => ∑ k, p k * y k) '' P)

def pyramid {n : ℕ} (q : Fin n → ℝ)
    (P : Set (Fin n → ℝ)) : Set (Fin n → ℝ) :=
  convexHull ℝ (insert q P)

def zonotope {n : ℕ} (a : Fin n → ℝ) (I : Finset ℕ)
    (g : ℕ → Fin n → ℝ) : Set (Fin n → ℝ) :=
  {p | ∃ t : ℕ → ℝ, (∀ i ∈ I, 0 ≤ t i ∧ t i ≤ 1) ∧
    p = a + ∑ i in I, t i • g i}

abbrev BHRSY.Ordering (n : ℕ) (v : Fin n) :=
  Fin (n - 1) ≃ {k : Fin n // k ≠ v}

def BHRSY.Ordering.pos {v : Fin n} (π : Ordering n v)
    (i : ℕ) : Fin n :=
  if h : i - 1 < n - 1 then (π ⟨i - 1, h⟩).1 else v

def BHRSY.hyperplane (v : Fin n) : Set (Fin n → ℝ) :=
  {p | p v = 0 ∧ ∑ k, p k = 1}

noncomputable def BHRSY.prefixSum {v : Fin n}
    (π : Ordering n v) (p : Fin n → ℝ) (i : ℕ) : ℝ :=
  ∑ j in Finset.Ico 1 i, p (π.pos j)

def BHRSY.parallelepiped (v : Fin n) (π : Ordering n v)
    (M : Finset ℕ) : Set (Fin n → ℝ) :=
  {p | p ∈ hyperplane v ∧
    (∀ i ∈ M, 0 ≤ prefixSum π p i ∧ prefixSum π p i ≤ 1) ∧
    ∀ i ∈ Finset.Icc 2 (n - 1), i ∉ M →
      -1 ≤ p (π.pos i) ∧ p (π.pos i) ≤ 0}

def BHRSY.paramid (v : Fin n) (π : Ordering n v)
    (M : Finset ℕ) : Set (Fin n → ℝ) :=
  pyramid (Pi.single v 1) (parallelepiped v π M)

noncomputable def BHRSY.paramidSupport (v : Fin n)
    (π : Ordering n v) (M : Finset ℕ) : (Fin n → ℝ) → ℝ :=
  supportFunction (paramid v π M)

def BHRSY.lastMarked (M : Finset ℕ) : ℕ :=
```

```
  (insert 1 M).max' (Finset.insert_nonempty _ _)

def BHRSY.markedParent (M : Finset ℕ) (i : ℕ) : ℕ :=
  (insert 1 (M.filter fun j => j < i)).max'
    (Finset.insert_nonempty _ _)
```

The dictionary uses the word $\pi' = (v, \pi_1, \ldots, \pi_{N-1})$ (`Ordering.word`), the cut set $S = M - 1$ (`cutSet`), and the description $(h_{\pi',S}, L_{\pi',S})$ of a word (`caterpillarWordDescription`). The rooted star is `Trees.rootedStar`. With $N = r + 2$, `Theorems.lean` states Theorem 14 as follows.

```
theorem bhrsy_parallelepiped_eq_zonotope {r : ℕ}
    {v : Fin (r + 1 + 1)} (π : BHRSY.Ordering (r + 1 + 1) v)
    {M : Finset ℕ} (hM : M ⊆ Finset.Icc 2 (r + 1)) :
    BHRSY.parallelepiped v π M =
      zonotope (Pi.single (π.pos (BHRSY.lastMarked M)) 1)
        (Finset.Icc 2 (r + 1))
        (fun i => Pi.single (π.pos (BHRSY.markedParent M i)) 1
          - Pi.single (π.pos i) 1)

theorem bhrsy_support_eq {r : ℕ} {v : Fin (r + 1 + 1)}
    (π : BHRSY.Ordering (r + 1 + 1) v) {M : Finset ℕ}
    (hM : M ⊆ Finset.Icc 2 (r + 1)) (y : Point (r + 1 + 1)) :
    BHRSY.paramidSupport v π M y =
      y v + MaxCaterpillar.relu
        (y (π.pos (BHRSY.lastMarked M)) - y v +
          ∑ i in Finset.Icc 2 (r + 1), MaxCaterpillar.relu
            (y (π.pos (BHRSY.markedParent M i)) - y (π.pos i)))

theorem bhrsy_support_eq_word {r : ℕ} {v : Fin (r + 1 + 1)}
    (π : BHRSY.Ordering (r + 1 + 1) v) {M : Finset ℕ}
    (hM : M ⊆ Finset.Icc 2 (r + 1)) :
    (BHRSY.cutSet (r := r) M).card = M.card ∧
      ∀ y : Point (r + 1 + 1), BHRSY.paramidSupport v π M y =
        y v + caterpillarFunction
          (caterpillarWordDescription π.word (BHRSY.cutSet M)) y

theorem bhrsy_support_mem_caterpillar {r : ℕ}
    {v : Fin (r + 1 + 1)} (π : BHRSY.Ordering (r + 1 + 1) v)
    {M : Finset ℕ} (hM : M ⊆ Finset.Icc 2 (r + 1))
    (hne : M.Nonempty) :
    ∃ k ∈ caterpillarDescriptions r, ∀ y,
      BHRSY.paramidSupport v π M y = y v + caterpillarFunction k y

theorem bhrsy_support_empty {r : ℕ} {v : Fin (r + 1 + 1)}
    (π : BHRSY.Ordering (r + 1 + 1) v) (y : Point (r + 1 + 1)) :
    BHRSY.paramidSupport v π ∅ y =
      y v + Trees.rootedStar v (π.pos 1) y

theorem rootedStar_ne_caterpillarFunction {r : ℕ}
```

```
{a c : Fin (r + 1 + 1)} (hac : a ≠ c) {k : FaceKey r}
(hk : k ∈ caterpillarDescriptions r) :
Trees.rootedStar a c ≠ caterpillarFunction k
```

The same file contains the converse `bhrsy_word_mem`, the special cases `bhrsy_parallelepiped_eq_zonotope_of_nonempty` and `bhrsy_parallelepiped_empty`, the statement `bhrsy_support_empty_not_caterpillar`, and the dichotomy `bhrsy_family`.

E.5 Version 1, 2, and 3 declarations

Improvements were added beside the existing development rather than replacing it. In the Lean sources, superseded declarations are kept and commented as Version 1 (and Version 2), and the new ones as Version 2 (and Version 3).

- *Main theorem.* Version 1, `maximum_width`, asserts a real-parameter network of the stated widths. Version 2, `maximum_width_rational`, adds rational weights and zero biases for the network compiled from the Version 1 coefficients. Version 3, `maximum_width_one_basis`, proves the same statement from one basis via `quotient_compression` and `quotientDim_add_two_le_secondWidth`; the public `caterpillar_net` restates it and is linked from Theorem 1.

- *Coefficients.* Version 1, `coefficient`, averages over all bases $S \subseteq \mathcal{D}$ of $\operatorname{span}\mathcal{D}$ the coordinates $G_S^{-1} b_S$ of $\max_N$, computed from the Gram matrix G_S and the vector b_S of inner products in $L^2([-1,1]^N)$; `coefficient_rational` proves it rational. It needs no choice of basis. Version 2, `rational_coefficients`, uses one basis and the evaluation argument in the proof of Theorem 11.

- *Compilation and widths.* Version 1 compiles the canonical coefficients only (`network`, `construction_width`). Version 2 compiles every expansion (`compilation`), and gives the widths of Theorems 1, 11 and 12.

The Version 1 and Version 2 statements remain checked.

The keyword `noncomputable` allows classical definitions of real quantities. The theorem is an exact representation statement; it does not by itself assert an efficient algorithm for extracting numerical weights.

E.6 Verification scope

The repository's verification certificate, dated 22 October 2026, records a build with Lean 4.4.0 and checks 102 public statements against their exact types. Its permitted axioms are `propext`, `Classical.choice`, and `Quot.sound`. The checked interface includes dictionary sufficiency, rational expansions, network correctness with rational weights and zero biases, the width bounds, the compilation and compression theorems, the small-arity networks, the Wang–Sun and Koutschan–Moser–Ponomarchuk–Schicho representations, and the network theorems for sums of maxima and CPWL functions with their rationality and bias statements. It covers every theorem, lemma, and numbered equation of the paper. The certificate and its source hashes are available in the linked verification record. The mathematical arguments are provided in the main text and Appendices A to C; the formalization is an additional certificate of the formal statements.

AI methodology statement

Claude Code and Codex were used for mathematical exploration and proof search, development of the Lean formalization, and drafting and revision of the manuscript. Their use materially affected the construction and proofs in Sections 3 to 8 and Appendices A to E. The author set the research objectives, directed the searches, and revised the mathematical presentation. The accompanying Lean certificate checks the formal statements and their axiom dependencies; it does not certify the prose exposition, literature comparison, or claims of originality. The author is responsible for the contents of the paper.

References

[ABMM18] Raman Arora, Amitabh Basu, Poorya Mianjy, and Anirbit Mukherjee. Understanding deep neural networks with rectified linear units. In *International Conference on Learning Representations*, 2018. arXiv:1611.01491v6.

[AHM25] Gennadiy Averkov, Christopher Hojny, and Maximilian Merkert. On the expressiveness of rational ReLU neural networks with bounded depth. In *International Conference on Learning Representations*, 2025. OpenReview.

[BBH+26] Egor Bakaev, Florestan Brunck, Christoph Hertrich, Jack Stade, and Amir Yehudayoff. Better neural network expressivity: Subdividing the simplex. In *Proceedings of the 58th Annual ACM Symposium on Theory of Computing*, pages 500–507, 2026. doi:10.1145/3798129.3800768.

[BHR+26] Florestan Brunck, Christoph Hertrich, Kilian Rueß, Jack Stade, and Amir Yehudayoff. Two ReLU layers is all you need. Preprint, arXiv:2610.28587, version 1, October 2026. arXiv:2610.28587v1.

[Fre42] Hans Freudenthal. Simplizialzerlegungen von beschränkter Flachheit. *Annals of Mathematics*, 43(3):580–582, 1942. doi:10.2307/1968813.

[GHL25] Moritz Grillo, Christoph Hertrich, and Georg Loho. Depth-bounds for neural networks via the braid arrangement. *Advances in Neural Information Processing Systems*, 38:127800–127818, 2025. doi:10.52202/085713-4257.

[HHL23] Christian Haase, Christoph Hertrich, and Georg Loho. Lower bounds on the depth of integral ReLU neural networks via lattice polytopes. In *International Conference on Learning Representations*, 2023. OpenReview.

[HBD+23] Christoph Hertrich, Amitabh Basu, Marco Di Summa, and Martin Skutella. Towards lower bounds on the depth of ReLU neural networks. *SIAM Journal on Discrete Mathematics*, 37(2):997–1029, 2023. doi:10.1137/22M1489332.

[KMPS23] Christoph Koutschan, Bernhard Moser, Anton Ponomarchuk, and Josef Schicho. Representing piecewise linear functions by functions with small arity. Preprint, arXiv:2305.16933, 2023. arXiv:2305.16933.

[Kuh60] Harold W. Kuhn. Some combinatorial lemmas in topology. *IBM Journal of Research and Development*, 4(5):518–524, 1960. doi:10.1147/rd.45.0518.

[Lu26] Shunqi Lu. Sharp closure thresholds for two-hidden-layer ReLU networks. Preprint, version 1, September 2026. doi:10.5281/zenodo.22728502. Author's PDF.

[RAB+26] Kilian Rueß, Gennadiy Averkov, Florestan Brunck, Moritz Grillo, Christoph Hertrich, Georg Loho, Jack Stade, Moritz Stargalla, Matthew Sun, and Martin Winter. Shallower ReLU network representations via exact linear algebra. Preprint, arXiv:2607.21651, version 2, September 2026. arXiv:2607.21651v2.

[Saf26] Itay Safran. A depth hierarchy for computing the maximum in ReLU networks via extremal graph theory. In *Proceedings of the 39th Conference on Learning Theory*, volume 336 of *Proceedings of Machine Learning Research*, pages 5721–5742, 2026. PMLR 336.

[WB26] Zhimao Wang and Amitabh Basu. Representing MAX functions using two-hidden-layer ReLU networks. Preprint, arXiv:2608.25221, August 2026. arXiv:2608.25221v1.

[WS05] Shuning Wang and Xusheng Sun. Generalization of hinging hyperplanes. *IEEE Transactions on Information Theory*, 51(12):4425–4431, 2005. doi:10.1109/TIT.2005.859246.